

NeuroHackademy 2026

The DIPY logo consists of the letters 'DIPY' in a large, orange, serif font with a black outline, set against a solid red rectangular background.

Overview of Diffusion imaging pipeline using DIPY

Serge KOUDORO

Intelligent Systems Engineering, Indiana University

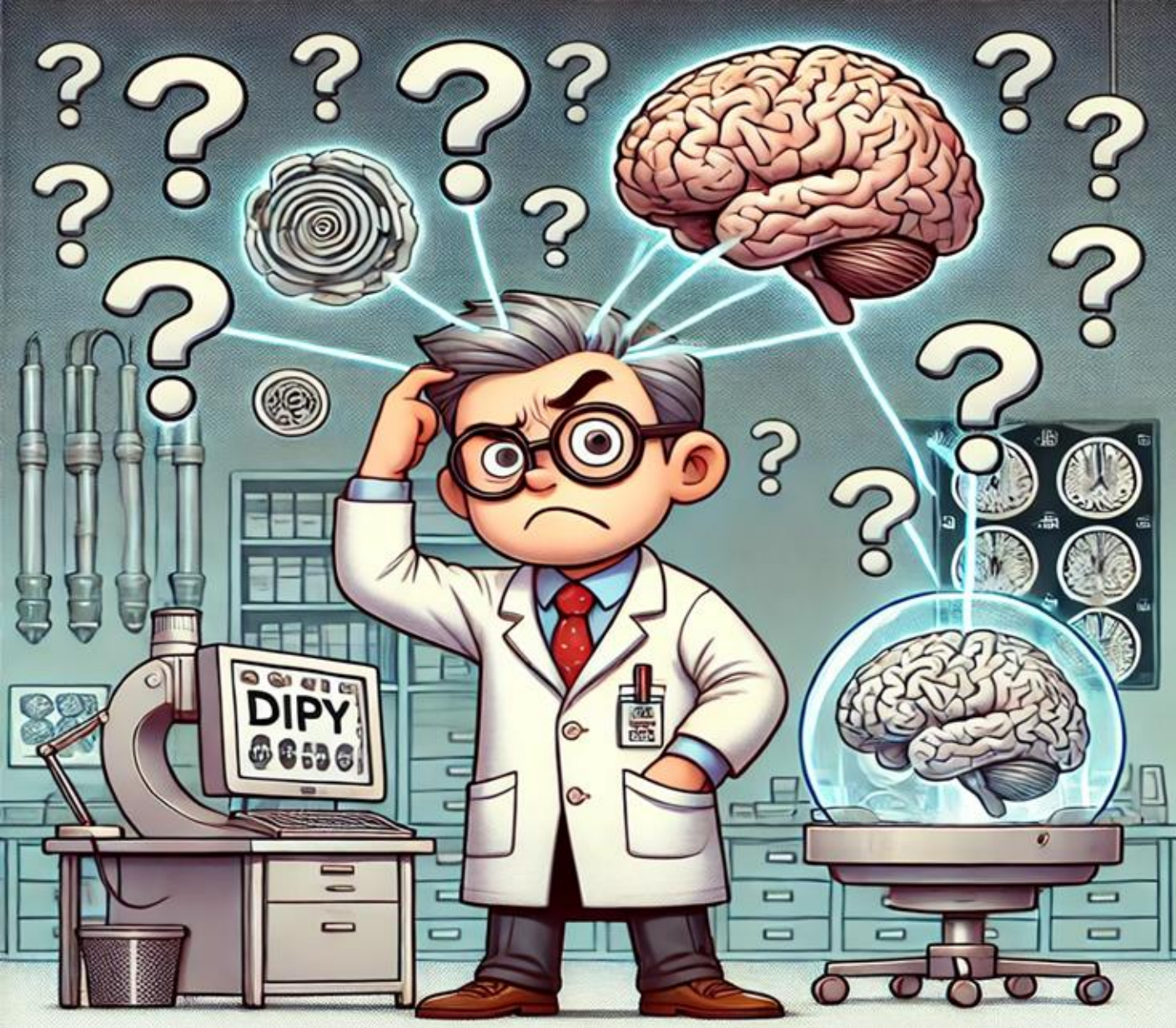
Key Learning Objectives

- ☐ Understand the fundamentals of diffusion MRI
- ☐ Explore DIPY and its command-line interface (CLI)
- ☐ Identify the essential blocks in a diffusion imaging pipeline
- ☐ Build your own diffusion imaging pipeline



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**Let's start
from the
beginning**



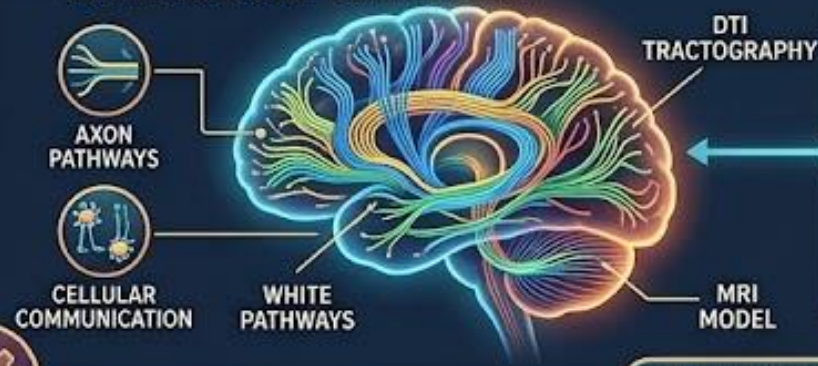
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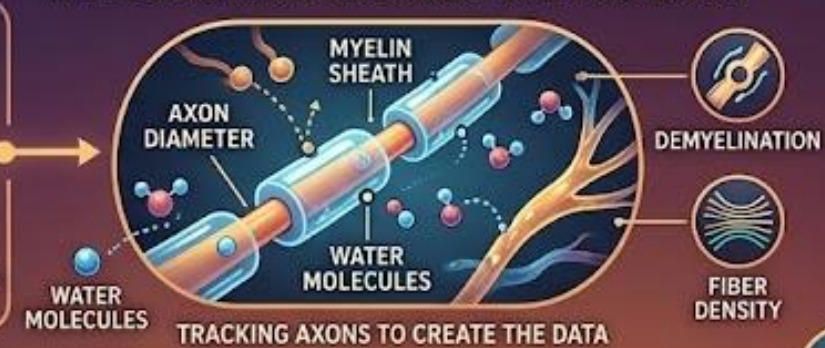
ULTIMATE GOALS: DIFFUSION MRI & NEUROSCIENCE

I. DIFFUSION MRI: NON-INVASIVE HISTOLOGY & THE CONNECTOME

A. MAPPING THE "CONNECTOME"

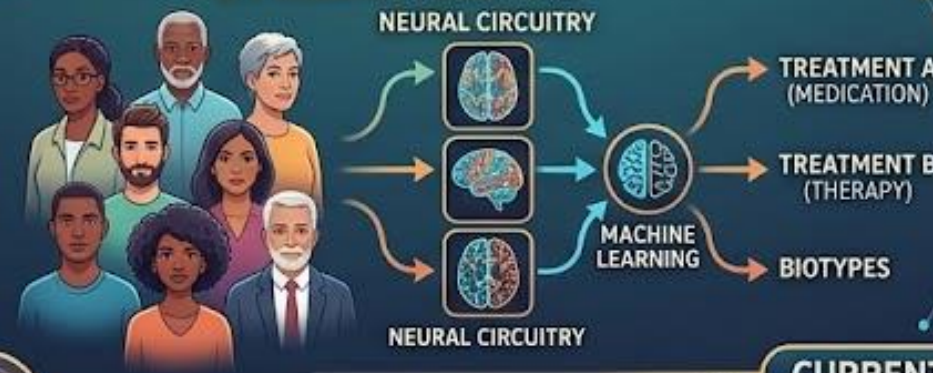


B. PROBING MICROSTRUCTURE & IN VIVO BIOPSY

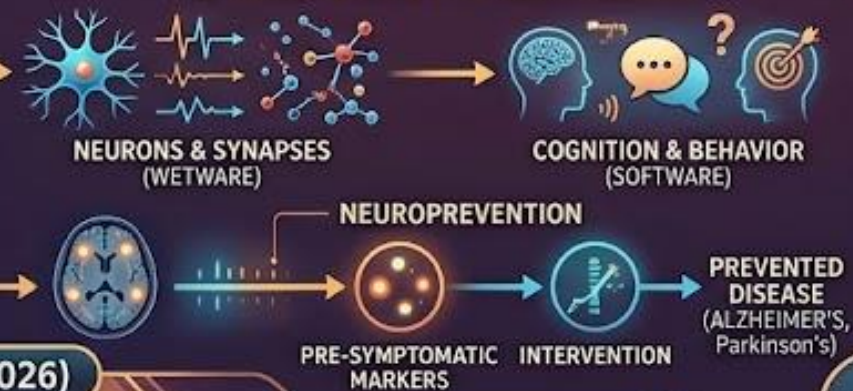


II. NEUROSCIENCE: THE "PREDICTIVE BRAIN"

C. PRECISION PSYCHIATRY



D. DECIPHERING THE NEURAL CODE



CURRENT FRONTIERS (2026)



1. SUB-MILLIMETER RESOLUTION
(CORTICAL LAYERS)



2. FAST SCANNING
(<5 MINS)



3. GENOMICS INTEGRATION
(DNA shaping brain)



4. PORTABLE
LOW-FIELD MRI



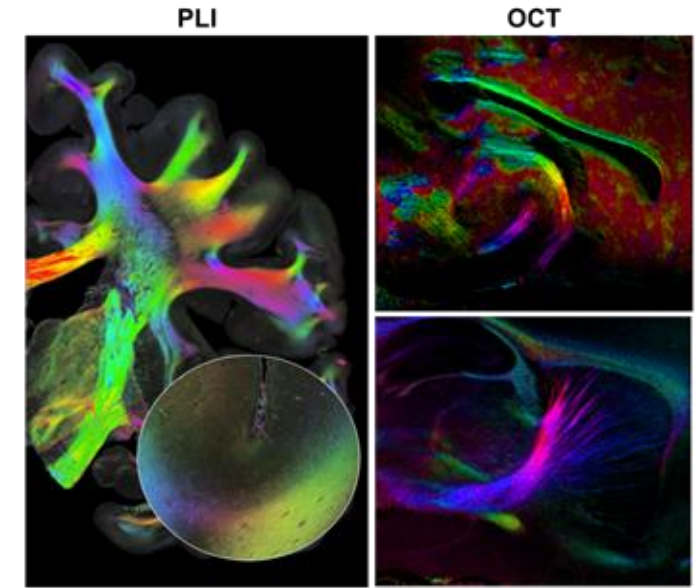
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How do we track connections in the human brain ?

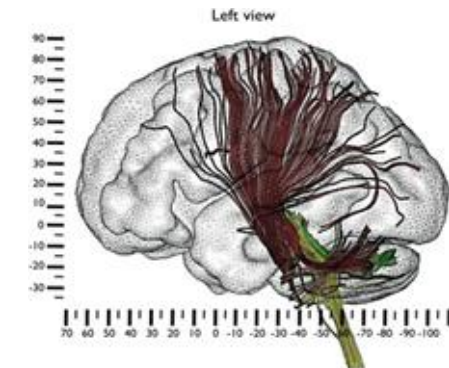
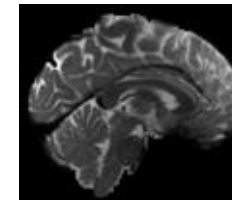
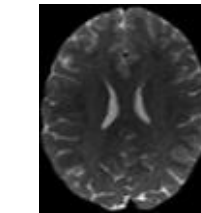
=> **Polarized light imaging (PLI) & Optical coherence tomography (OCT):**

- First developed in the 1990s
- Postmortem
- Resolution: micrometer
- PLI: birefringent (reflective) properties of myelinated fibers
- OCT: interference of light backscattered from the tissue



=> **Diffusion Magnetic Resonance Imaging (dMRI):**

- First developed in the 1990s
- *in vivo*
- Resolution: millimeter
- anisotropic diffusivity of water in the axons



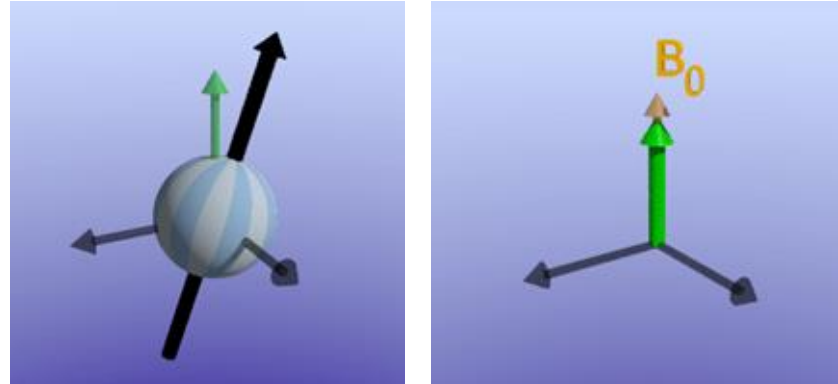
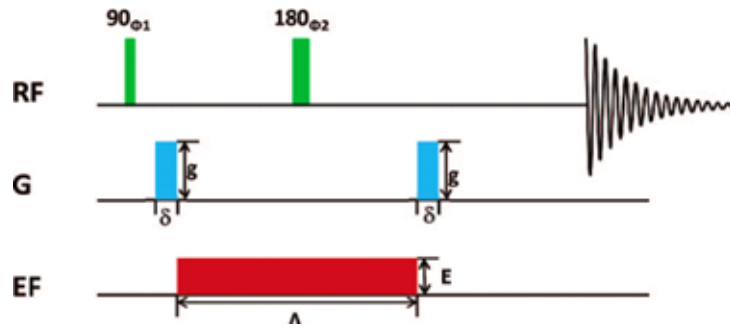
Catani et al., *Cortex*,
2008



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Pulsed-gradient spin echo experiment



Stejskal and Tanner (1965)

$$\frac{S(TE, \gamma, G, \Delta, \delta)}{S(TE)} = e^{-\gamma^2 G^2 \delta^2 (\Delta - \frac{\delta}{3}) D}$$

$$\frac{S(b)}{S_0} = e^{-bD}$$

MR diffusion tensor spectroscopy and imaging

P J Basser¹, J Mattiello, D LeBihan

Affiliations + expand

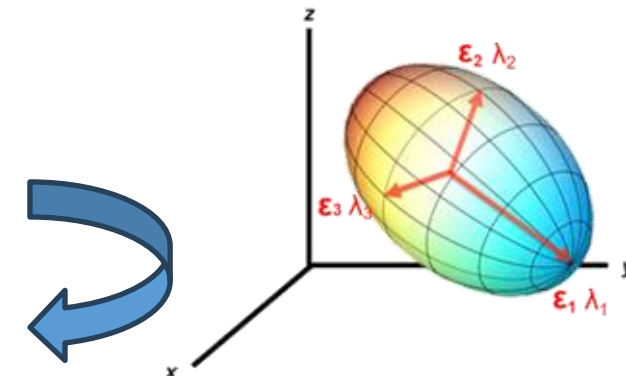
PMID: 8130344 PMCID: PMC1275686 DOI: 10.1016/S0006-3495(94)80775-1

$$\ln \left(\frac{S(\theta)}{S_0} \right) = -b \mathbf{g}^T \mathbf{D} \mathbf{g}$$



Fitted as a linear model → D

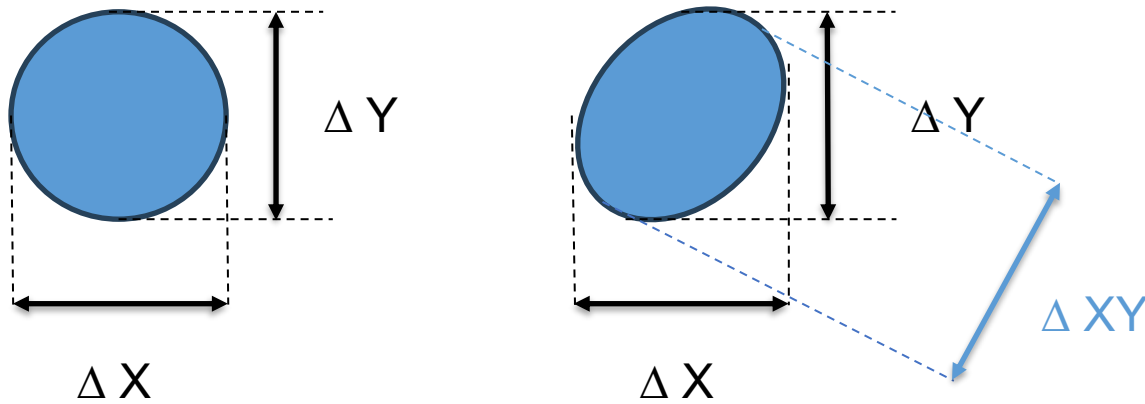
Eigen decomposition



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Why do we need tensor representation?



$$\begin{bmatrix} D_{xx} & D_{xy} \\ D_{yx} & D_{yy} \end{bmatrix}$$

10

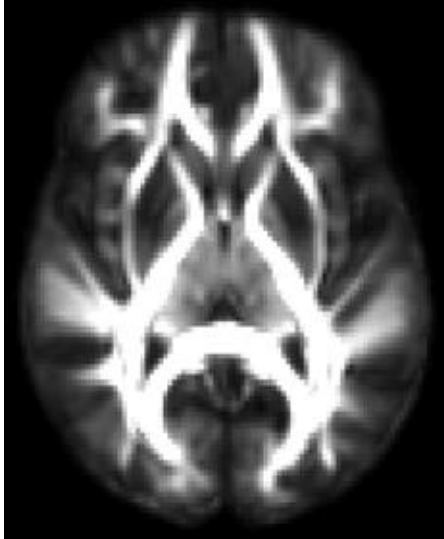


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10

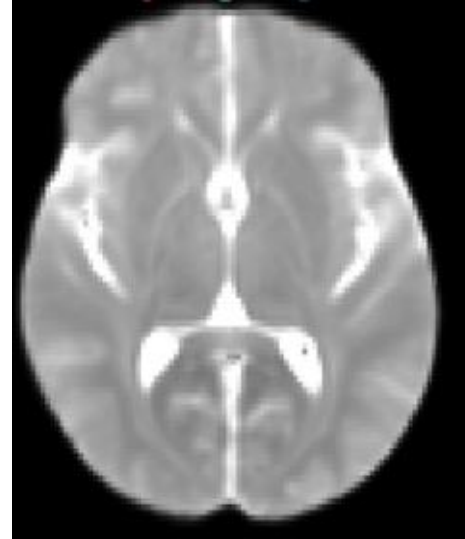
FA map



$$FA = \sqrt{\frac{3}{2}} \cdot \frac{\sqrt{(\lambda_1 - \lambda_{mean})^2 + (\lambda_2 - \lambda_{mean})^2 + (\lambda_3 - \lambda_{mean})^2}}{\sqrt{\lambda_1^2 + \lambda_2^2 + \lambda_3^2}}$$

- Useful for detecting neuro-degeneracies
- Useful for detecting tracts and their connectivity patterns
- Assess white matter integrity

MD map



$$MD = \frac{\lambda_1 + \lambda_2 + \lambda_3}{3}$$

- Detect tissue changes
- Useful for analyzing edema
- Assess different pathologies and axonal density

Software

- FSL (2000)
- Camino (2005)
- Tortoise (2007)
- Mrtrix1 (2009) , Mrtrix3 (2019)
- Explore-DTI (2009)
- DIPY (2011)



DIPY

Free and open source software project for computational anatomy in Python.

It focuses on diffusion MRI analysis but also contains algorithms that are generic for medical imaging e.g. denoising, registration.

Created in 2011, DIPY is an international project which brings together scientists **across labs and countries ...**

... to share their **state-of-the-art code** and expertise in the same codebase, accelerating scientific research in medical imaging.

Specialized: **Diffusion MRI**

- ☐ Data preprocessing (P2S, Gibbs, ...)
- ☐ Voxel signal reconstruction (DKI, QTI, ...)
- ☐ Tracking algorithms (PFT, ...)
- ☐ Tractography Segmentation (QB, RB, ...)
- ☐ Tractography Registration (SLR, BundleWarp, ...)
- ☐ Statistical analysis and tractometry (BUAN, AFQ, ...).

General: **Medical Imaging**

- ☐ Affine and Nonrigid registration (SyN, ...)
- ☐ Denoising (NLMeans, ...)
- ☐ Visualization (Horizon, FURY, ...)
- ☐ Tissue segmentation & classification (HMRF, ...)

Garyfallidis, Eleftherios, Matthew Brett, Bagrat Amirbekian, Ariel Rokem, Stefan Van Der Walt, Maxime Descoteaux, and Ian Nimmo-Smith. "**Dipy, a library for the analysis of diffusion MRI data.**" Frontiers in neuroinformatics 8 (2014): 8.

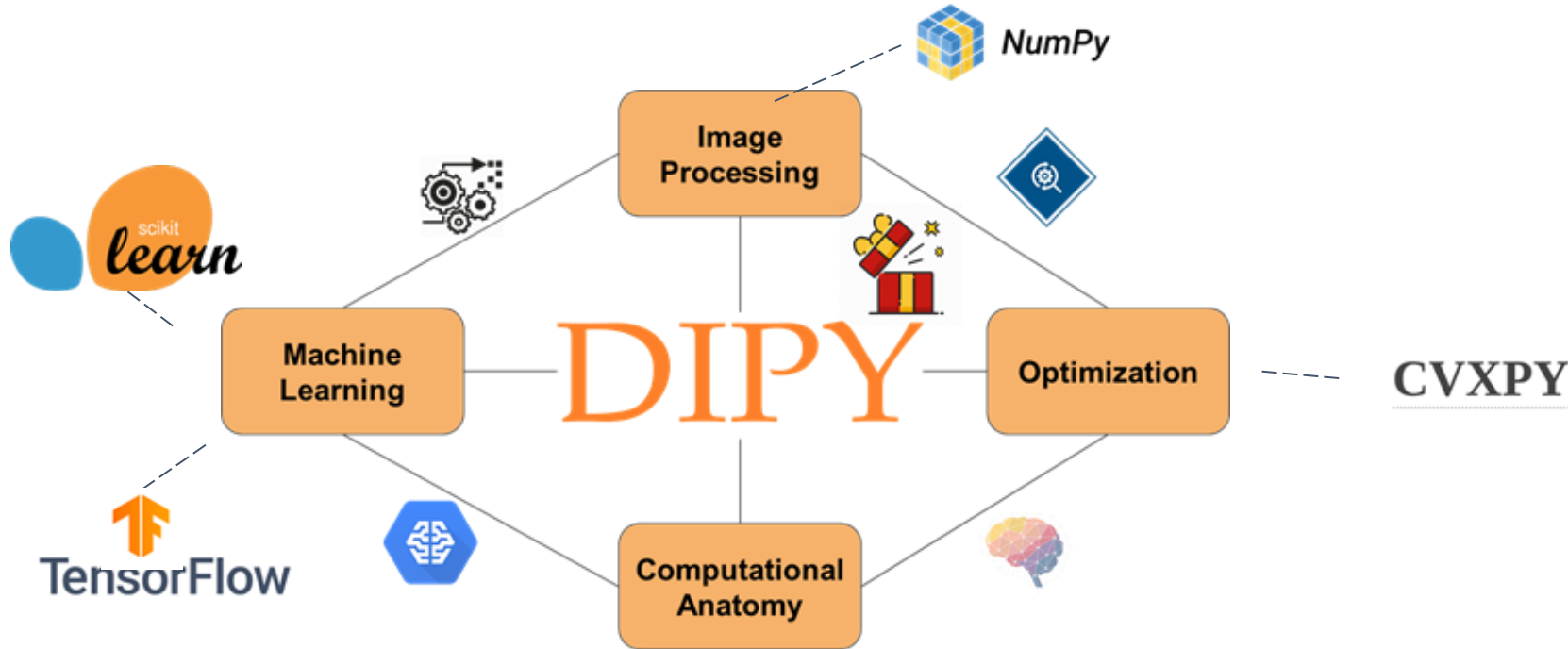


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DIPY Dependencies & Ecosystem

DIPY has been developed in parallel with NiBabel to provide access to Pythonic APIs for Neuroimagers and Data Scientists.



A Growing DIPY Ecosystem

- ☐ GPUStreamlines (NVIDIA)
- ☐ cuDIPY, FURY, BrainLife
- ☐ dsimpy
- ☐ dmipy, pyAFQ, preQual
- ☐ Ndmg, QSIprep, Tractoflow
- ☐ MNE Python ...
- ☐ Used by 689+ public projects
- ☐ Used in industry and clinic for more than 10 years.
- ☐ 20+ medical startups use DIPY.

Garyfallidis, Eleftherios, Matthew Brett, Bagrat Amirbekian, Ariel Rokem, Stefan Van Der Walt, Maxime Descoteaux, and Ian Nimmo-Smith. "Dipy, a library for the analysis of diffusion MRI data." *Frontiers in neuroinformatics* 8 (2014): 8.



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Growing the TRL (Technology Readiness Level)

TRL	Software-Specific Definition	Indicators	What Does it means for DIPY ?
TRL-1	Basic research and concept formulation. No code exists yet.	Theoretical models, academic papers, white papers, proof-of-concept ideas.	Literature review
TRL-2	Algorithm development and feasibility studies. No practical implementation.	Early algorithm descriptions, feasibility tests, and simulations.	Algorithm Development: Core diffusion MRI methods are based on well-established academic research.
TRL-3	Prototype or preliminary code implemented in a controlled, test environment.	Alpha version of software, preliminary validation using synthetic data.	
TRL-4	Functional prototype with basic usability. Demonstrated in a lab setting, not tested externally.	First public release, early validation against real-world data, but limited users.	Code Implementation: Open-source implementation with robust Python support. - Strong Software Engineering principle + core API vs UI - Heavy Testing + Detail documentation +Community - Great and sustainable software to help researchers.
TRL-5	Software validated in a relevant environment (e.g., used in research).	Beta version, limited external users, integrated with test datasets, early debugging.	

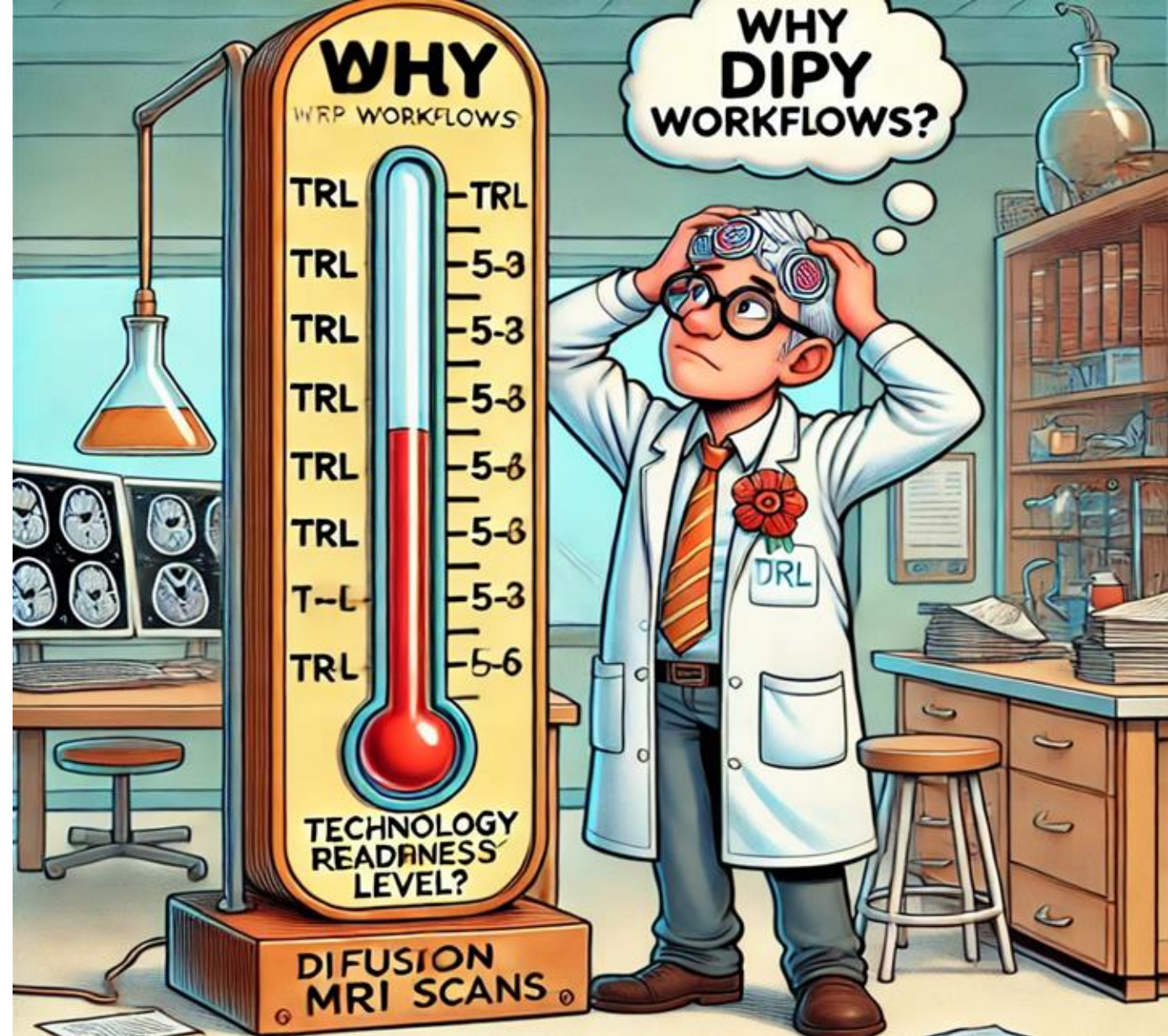


Growing the TRL (Technology Readiness Level)

TRL	Software-Specific Definition	Indicators	What Does it means for DIPY ?
TRL-6	Software tested in a real-world research setting, but not fully adopted or standardized.	Multiple research groups use it, academic publications cite it, early real-world applications.	Research Validation: Used in academic research, validated in studies.
TRL-7	Operational prototype demonstrated in a real environment, but not yet a commercial or clinical standard.	Widespread academic use, reliable documentation, community contributions, open-source adoption.	Operational Deployment: Widely used in neuroscience, but limited clinical deployment.
TRL-8	Software is fully validated and used in production, but not yet a formal regulatory or enterprise standard.	Software integrated into commercial or clinical systems, tested on a broad scale.	Industry Adoption & Regulatory Compliance: No FDA/CE approval yet for medical diagnostics, not embedded in commercial MRI systems.
TRL-9	Fully deployed in operational environments, meeting industry or regulatory standards.	Regulatory approval (e.g., FDA, ISO), integration into commercial software suites.	Regulatory Approval: FDA clearance or CE marking - Enterprise Integration - Long-Term Clinical Validation - Security & Compliance - Broader Adoption Beyond Academia:

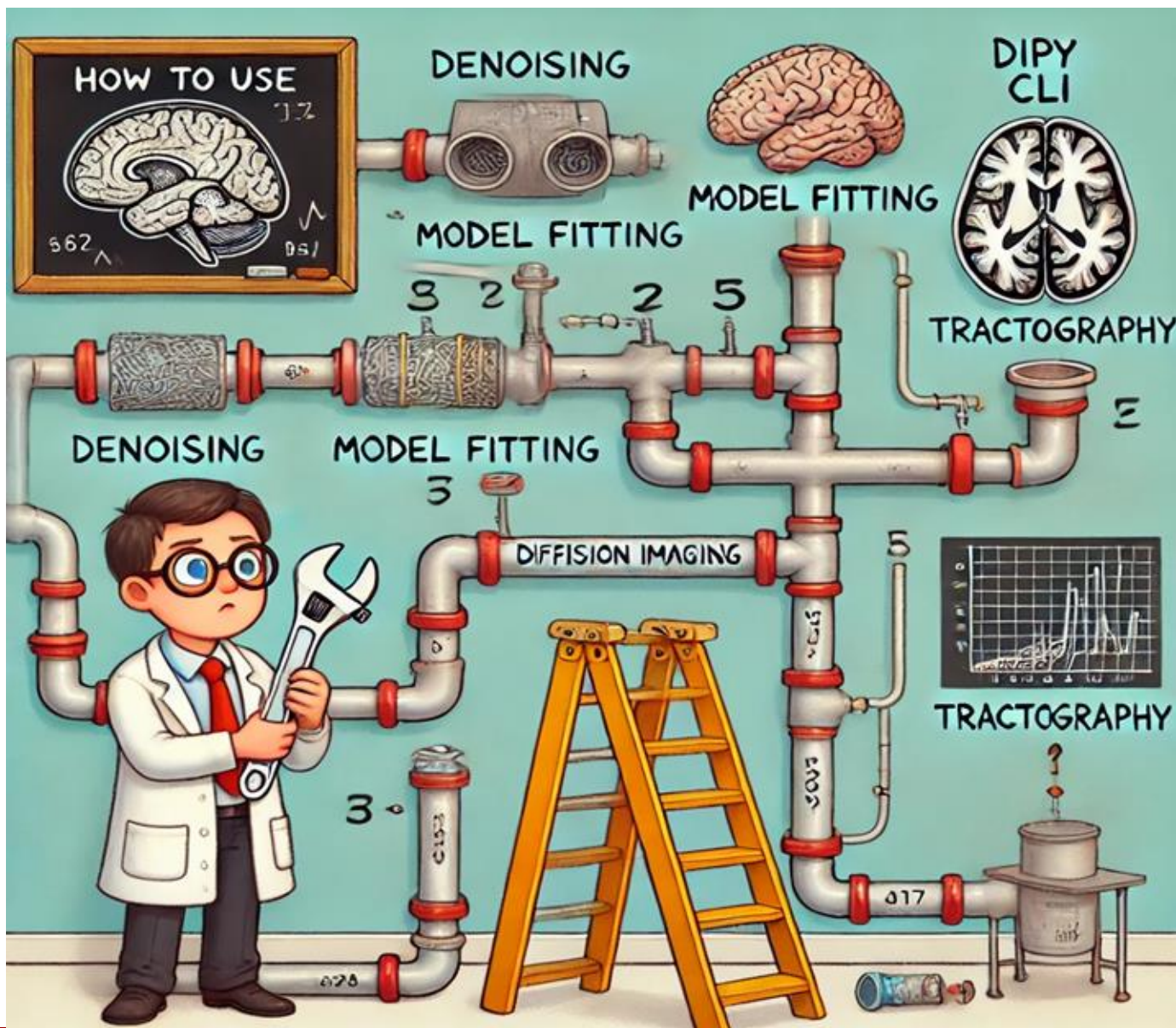
➔ TR-7: Operational Deployment

- Reduces manual effort by enabling automation and scripting
- Accessible to non-programmers
- Enhances reproducibility
- Scale efficiently for large datasets / Optimized for efficiency and scalability



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**What are the
components
of a
dMRI pipeline
?**



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First things first, dcm2nix => DICOM to NIFTI

`dcm2niix -z y -f %p_%t_%s -o /path/output /path/to/dicom/folder`

```
[boystown] [p10PM] /nobackup/p10PM/boystown/ASR0_Protocol/M68103049/t1w_32ch_mpr_1mm_00051.s
SeriesMR-0005-0001-1.dcm SeriesMR-0005-0015-1.dcm SeriesMR-0005-0029-1.dcm SeriesMR-0005-0043-1.dcm SeriesMR-0005-0057-1.dcm SeriesMR-0005-0071-1.dcm SeriesMR-0005-0085-1.dcm SeriesMR-0005-0099-1.dcm SeriesMR-0005-0113-1.dcm SeriesMR-0005-0127-1.dcm SeriesMR-0005-0141-1.dcm SeriesMR-0005-0155-1.dcm SeriesMR-0005-0169-1.dcm SeriesMR-0005-0183-1.dcm
SeriesMR-0005-0002-1.dcm SeriesMR-0005-0016-1.dcm SeriesMR-0005-0030-1.dcm SeriesMR-0005-0044-1.dcm SeriesMR-0005-0058-1.dcm SeriesMR-0005-0072-1.dcm SeriesMR-0005-0086-1.dcm SeriesMR-0005-0100-1.dcm SeriesMR-0005-0114-1.dcm SeriesMR-0005-0128-1.dcm SeriesMR-0005-0142-1.dcm SeriesMR-0005-0156-1.dcm SeriesMR-0005-0170-1.dcm SeriesMR-0005-0184-1.dcm
SeriesMR-0005-0003-1.dcm SeriesMR-0005-0017-1.dcm SeriesMR-0005-0031-1.dcm SeriesMR-0005-0045-1.dcm SeriesMR-0005-0059-1.dcm SeriesMR-0005-0073-1.dcm SeriesMR-0005-0087-1.dcm SeriesMR-0005-0101-1.dcm SeriesMR-0005-0115-1.dcm SeriesMR-0005-0129-1.dcm SeriesMR-0005-0143-1.dcm SeriesMR-0005-0157-1.dcm SeriesMR-0005-0171-1.dcm SeriesMR-0005-0185-1.dcm
SeriesMR-0005-0004-1.dcm SeriesMR-0005-0018-1.dcm SeriesMR-0005-0032-1.dcm SeriesMR-0005-0046-1.dcm SeriesMR-0005-0060-1.dcm SeriesMR-0005-0074-1.dcm SeriesMR-0005-0088-1.dcm SeriesMR-0005-0102-1.dcm SeriesMR-0005-0116-1.dcm SeriesMR-0005-0130-1.dcm SeriesMR-0005-0144-1.dcm SeriesMR-0005-0158-1.dcm SeriesMR-0005-0172-1.dcm SeriesMR-0005-0186-1.dcm
SeriesMR-0005-0005-1.dcm SeriesMR-0005-0019-1.dcm SeriesMR-0005-0033-1.dcm SeriesMR-0005-0047-1.dcm SeriesMR-0005-0061-1.dcm SeriesMR-0005-0075-1.dcm SeriesMR-0005-0089-1.dcm SeriesMR-0005-0103-1.dcm SeriesMR-0005-0117-1.dcm SeriesMR-0005-0131-1.dcm SeriesMR-0005-0145-1.dcm SeriesMR-0005-0159-1.dcm SeriesMR-0005-0173-1.dcm SeriesMR-0005-0187-1.dcm
SeriesMR-0005-0006-1.dcm SeriesMR-0005-0020-1.dcm SeriesMR-0005-0034-1.dcm SeriesMR-0005-0048-1.dcm SeriesMR-0005-0062-1.dcm SeriesMR-0005-0076-1.dcm SeriesMR-0005-0090-1.dcm SeriesMR-0005-0104-1.dcm SeriesMR-0005-0118-1.dcm SeriesMR-0005-0132-1.dcm SeriesMR-0005-0146-1.dcm SeriesMR-0005-0160-1.dcm SeriesMR-0005-0174-1.dcm SeriesMR-0005-0188-1.dcm
SeriesMR-0005-0007-1.dcm SeriesMR-0005-0021-1.dcm SeriesMR-0005-0035-1.dcm SeriesMR-0005-0049-1.dcm SeriesMR-0005-0063-1.dcm SeriesMR-0005-0077-1.dcm SeriesMR-0005-0091-1.dcm SeriesMR-0005-0105-1.dcm SeriesMR-0005-0119-1.dcm SeriesMR-0005-0133-1.dcm SeriesMR-0005-0147-1.dcm SeriesMR-0005-0161-1.dcm SeriesMR-0005-0175-1.dcm SeriesMR-0005-0189-1.dcm
SeriesMR-0005-0008-1.dcm SeriesMR-0005-0022-1.dcm SeriesMR-0005-0036-1.dcm SeriesMR-0005-0050-1.dcm SeriesMR-0005-0064-1.dcm SeriesMR-0005-0078-1.dcm SeriesMR-0005-0092-1.dcm SeriesMR-0005-0106-1.dcm SeriesMR-0005-0120-1.dcm SeriesMR-0005-0134-1.dcm SeriesMR-0005-0148-1.dcm SeriesMR-0005-0162-1.dcm SeriesMR-0005-0176-1.dcm SeriesMR-0005-0190-1.dcm
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SeriesMR-0005-0010-1.dcm SeriesMR-0005-0024-1.dcm SeriesMR-0005-0038-1.dcm SeriesMR-0005-0052-1.dcm SeriesMR-0005-0066-1.dcm SeriesMR-0005-0080-1.dcm SeriesMR-0005-0094-1.dcm SeriesMR-0005-0108-1.dcm SeriesMR-0005-0122-1.dcm SeriesMR-0005-0136-1.dcm SeriesMR-0005-0150-1.dcm SeriesMR-0005-0164-1.dcm SeriesMR-0005-0178-1.dcm SeriesMR-0005-0192-1.dcm
SeriesMR-0005-0011-1.dcm SeriesMR-0005-0025-1.dcm SeriesMR-0005-0039-1.dcm SeriesMR-0005-0053-1.dcm SeriesMR-0005-0067-1.dcm SeriesMR-0005-0081-1.dcm SeriesMR-0005-0095-1.dcm SeriesMR-0005-0109-1.dcm SeriesMR-0005-0123-1.dcm SeriesMR-0005-0137-1.dcm SeriesMR-0005-0151-1.dcm SeriesMR-0005-0165-1.dcm SeriesMR-0005-0179-1.dcm
SeriesMR-0005-0012-1.dcm SeriesMR-0005-0026-1.dcm SeriesMR-0005-0040-1.dcm SeriesMR-0005-0054-1.dcm SeriesMR-0005-0068-1.dcm SeriesMR-0005-0082-1.dcm SeriesMR-0005-0096-1.dcm SeriesMR-0005-0110-1.dcm SeriesMR-0005-0124-1.dcm SeriesMR-0005-0138-1.dcm SeriesMR-0005-0152-1.dcm SeriesMR-0005-0166-1.dcm SeriesMR-0005-0180-1.dcm
SeriesMR-0005-0013-1.dcm SeriesMR-0005-0027-1.dcm SeriesMR-0005-0041-1.dcm SeriesMR-0005-0055-1.dcm SeriesMR-0005-0069-1.dcm SeriesMR-0005-0083-1.dcm SeriesMR-0005-0097-1.dcm SeriesMR-0005-0111-1.dcm SeriesMR-0005-0125-1.dcm SeriesMR-0005-0139-1.dcm SeriesMR-0005-0153-1.dcm SeriesMR-0005-0167-1.dcm SeriesMR-0005-0181-1.dcm
SeriesMR-0005-0014-1.dcm SeriesMR-0005-0028-1.dcm SeriesMR-0005-0042-1.dcm SeriesMR-0005-0056-1.dcm SeriesMR-0005-0070-1.dcm SeriesMR-0005-0084-1.dcm SeriesMR-0005-0098-1.dcm SeriesMR-0005-0112-1.dcm SeriesMR-0005-0126-1.dcm SeriesMR-0005-0140-1.dcm SeriesMR-0005-0154-1.dcm SeriesMR-0005-0168-1.dcm SeriesMR-0005-0182-1.dcm
```



```
M68103049_t1w_32ch_mpr_1mm_20240501174427_5.json
M68103049_t1w_32ch_mpr_1mm_20240501174427_5.nii.gz
```

Li X, Morgan PS, Ashburner J, Smith J, Rorden C (2016) The first step for neuroimaging data analysis: DICOM to NiftI conversion. J Neurosci Methods. 264:47-56. doi: 10.1016/j.jneumeth.2016.03.001. [PMID: 26945974](https://pubmed.ncbi.nlm.nih.gov/26945974/)



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Installation

Official latest stable

```
>>> pip install -U dipy
```

```
>>> conda -c conda-forge dipy
```

Official developer / unstable

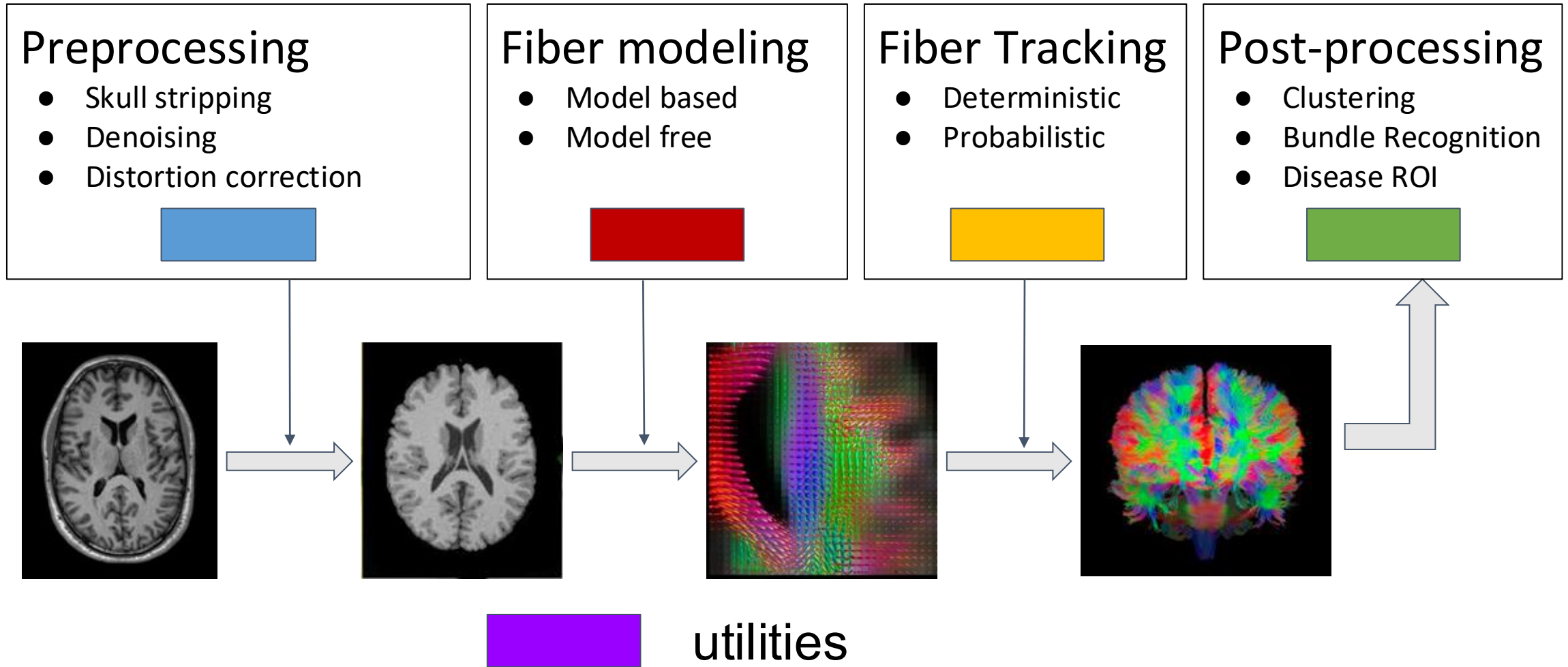
```
>>> pip install --upgrade --progress-bar off --extra-index-url  
"https://pypi.anaconda.org/scientific-python-nightly-  
wheels/simple" "dipy==1.12.0.dev0"
```



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CLI Diffusion MRI pipeline



➔ Currently, 50+ DIPY Workflows for diffusion MRI analysis. More to come....



 Preprocessing  Tractometry  utilities

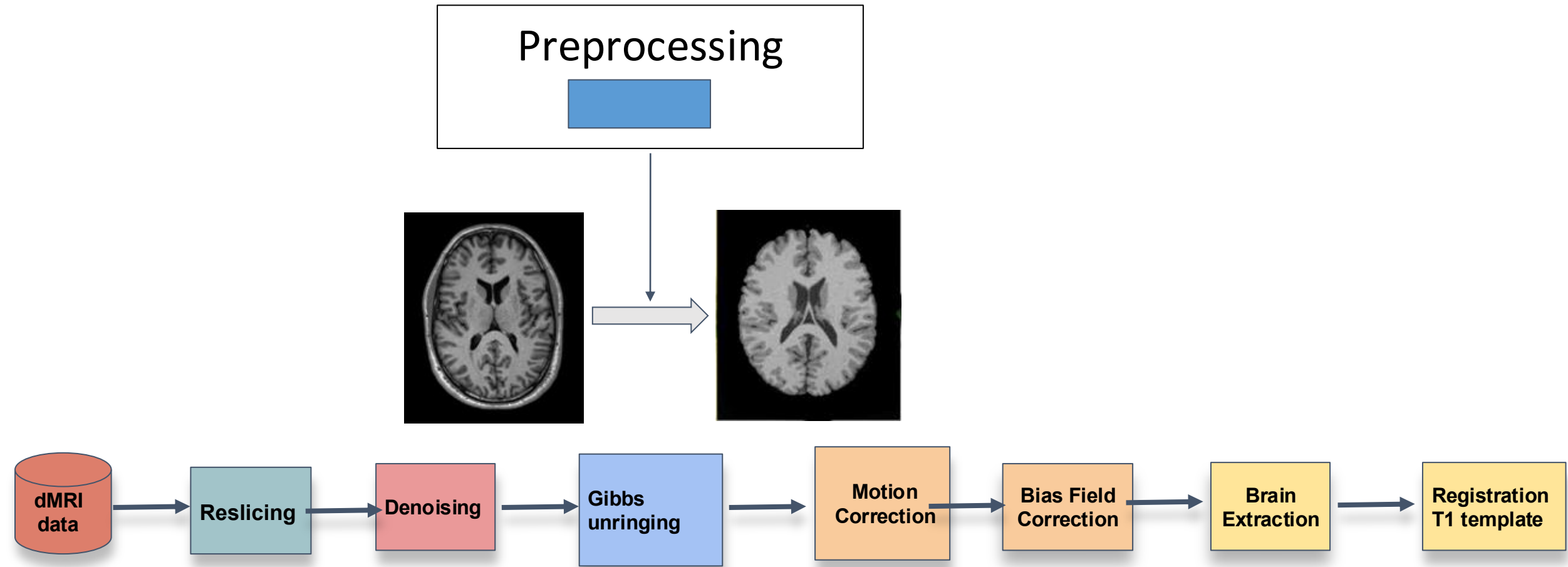
 Reconstruction  Tractography



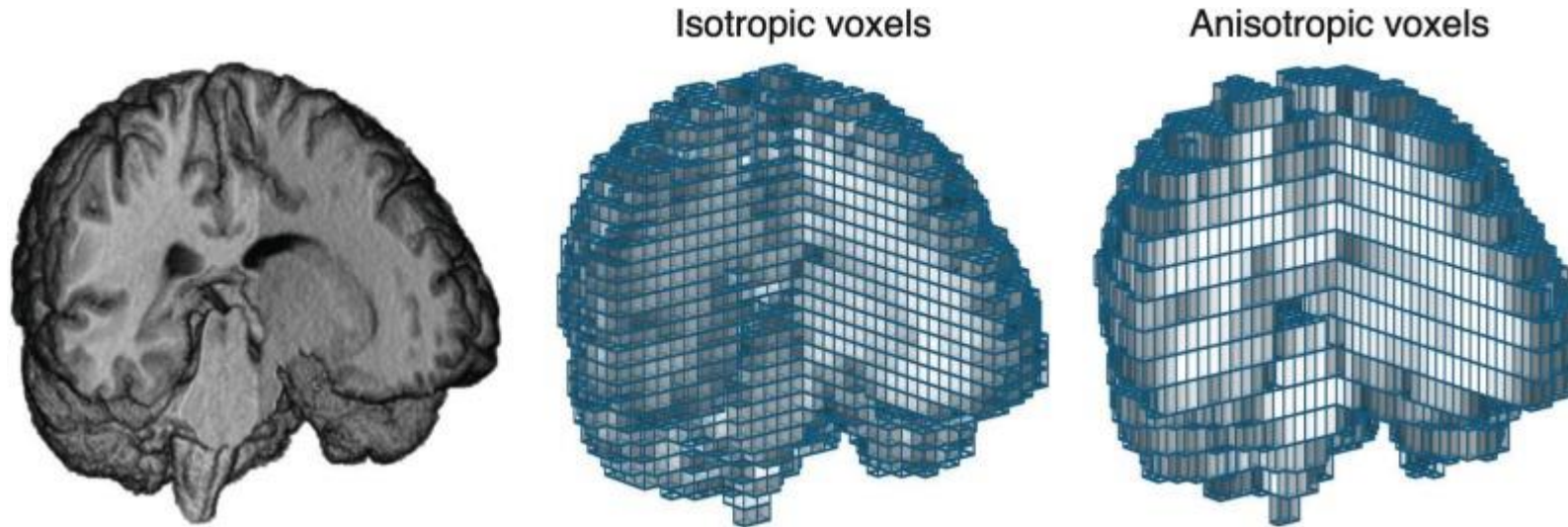
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dMRI pipeline: Why Preprocessing ?



Reslice (Interpolate)



```
>>> dipy_info my_nifiti_files.nii.gz  
>>> dipy_reslice
```



➡ Understand Your Data: >>> dipy_info

```
> dipy_info ../../dipy/stanford_hardi/HARDI150.nii.gz
INFO:-----
INFO:Looking at ../../dipy/stanford_hardi/HARDI150.nii.gz
INFO:-----
INFO:Data size (81, 106, 76, 160)
INFO:Data type int16
INFO:Data min 0 max 13157 avg 330.8058252724754 of vol 0
INFO:2nd percentile 0.0 98th percentile 2202.0 of vol 0
INFO:Native coordinate system RAS
INFO:Affine Native to RAS matrix
[[ 2.  0.  0. -80.]
 [ 0.  2.  0. -120.]
 [ 0.  0.  2. -60.]
 [ 0.  0.  0.  1.]]
INFO:Voxel size [2. 2. 2.]
```



➡ Understand Your Data: >>> dipy_info

```
> dipy_info ../../data/sunita/Dipy/abcdefg.nii.gz
INFO:-----
INFO:Looking at ../../data/sunita/Dipy/abcdefg.nii.gz
INFO:-----
INFO:Data size (128, 128, 88, 33)
INFO:Data type float64
INFO:Data min 0.0 max 25449.873025417328 avg 715.9515434390219 of vol 0
INFO:2nd percentile 0.0 98th percentile 5394.503000259399 of vol 0
INFO:Native coordinate system LAS
INFO:Affine Native to RAS matrix
[[ -1.868  -0.147  -0.057 121.132]
 [ -0.152   1.862   0.174 -78.635]
 [ -0.04   -0.167   1.992 -99.582]
 [  0.      0.      0.      1.    ]]
INFO:Voxel size [1.875 1.875 2. ]
WARNING:Voxel size is not isotropic. Please reslice.
```



Denoising

```
>>> dipy_denoise_nlmeans
```

Self-Similarity:

Images are often self-similar, in that each patch in an image is similar to many other patches from the same image.

Image denoising with **block-matching and 3D filtering** (Dabov, 2006)

NL-Means (Buades et. al. 2005) and its variants ONLM, AONLM, etc.

BM3D (Dabov et al., 2007) thresholds in frequency space.

Not implemented

Sparsity:

Images are often sparse and this property can be leveraged for denoising the data. These methods require a predetermined basis set to do the denoising such as DCT, DWT, etc. or by learning the basis from the data, known as dictionary learning.

K-SVD Overcomplete Dictionary Learning - Aarhon and Elad 2006

Denoising and fast diffusion imaging with physically constrained sparse dictionary learning, Gramfort, et. al., 2014

```
>>> dipy_denoise_lpca
>>> dipy_denoise_mppca
```

Low Rank Approximations:

Approximate the data to a smaller basis, typically via a low-rank approximation and then reconstruct from the low rank approximations.

Rank 'r' is the parameter that needs to be found for doing such an approximation. Typically done via PCA for Diffusion MRI.

Local PCA (Manjon et. al., 2013)

Marchenko Pastur PCA (Veraart et. al., 2016)

```
>>> dipy_denoise_patch2self
```

Statistical Independence:

Since noise originates from random fluctuations, noise in one subset of an image can be assumed to be statistically independent of the noise in another subset. Using one subset to predict the other gives denoising performance.

Patch2Self, Fadnavis, et. al. 2020

Noise2Self, Batson, et. al. 2019

Noise2Noise, Lehtinen, et. al. 2018

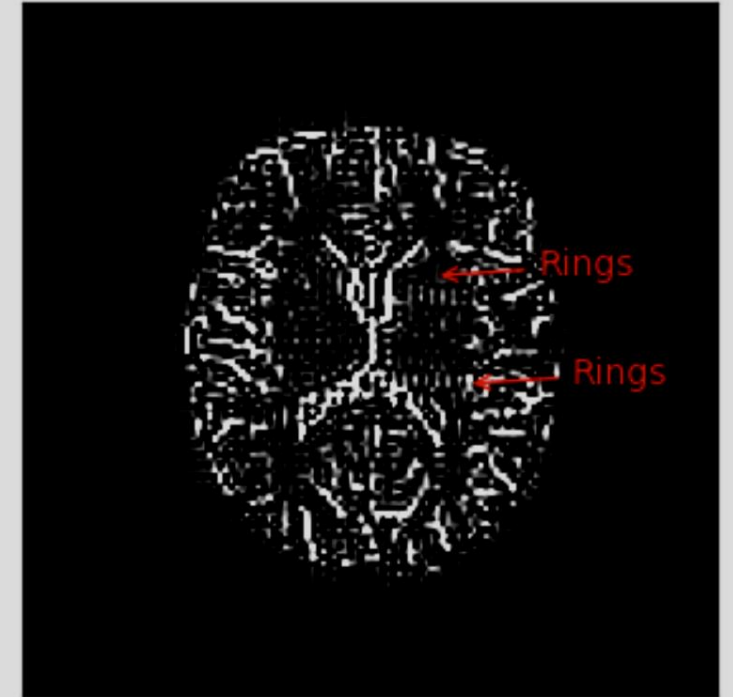
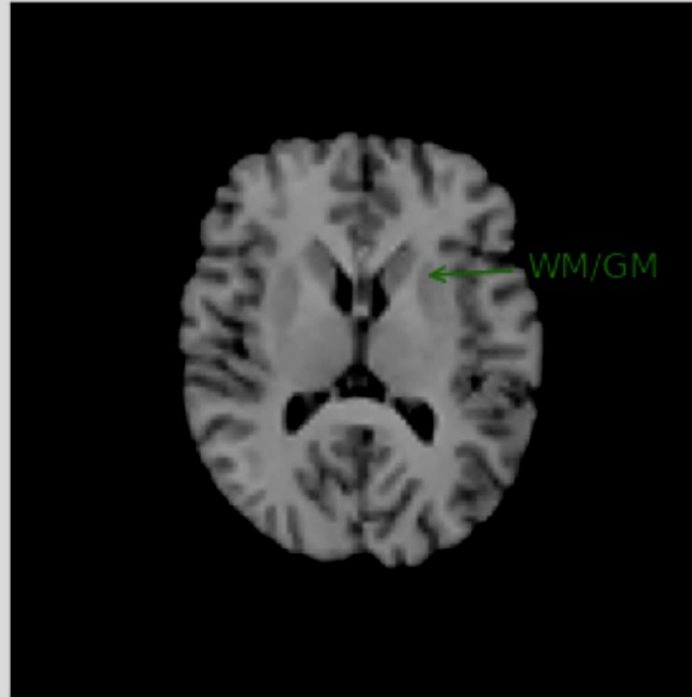
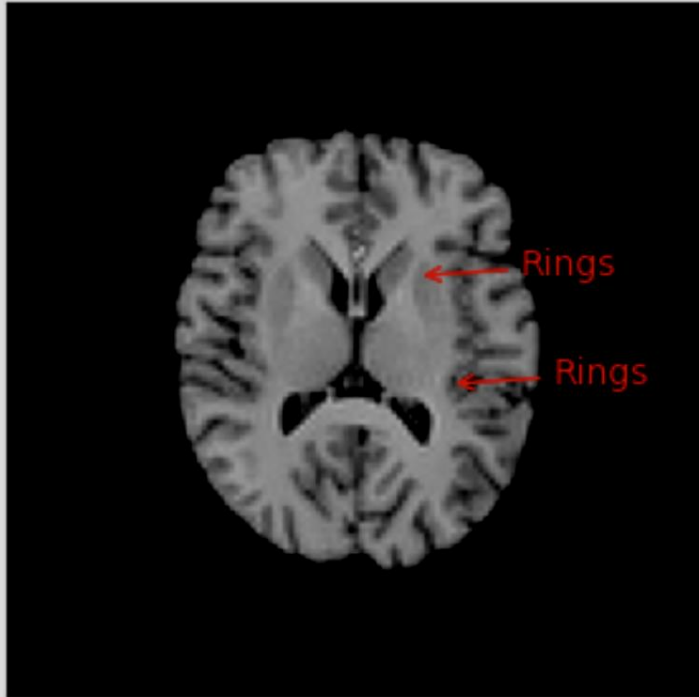


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Gibbs Unringing

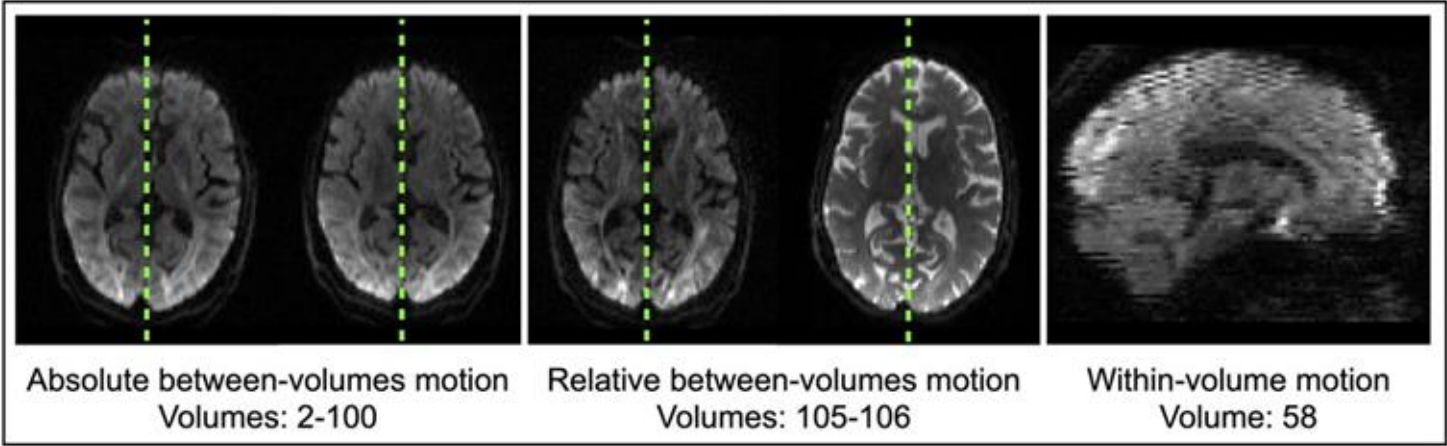
```
>>> dipy_gibbs_unringing input_files
```



Motion Correction

>>> dipy_correct_motion

Eddy current correction



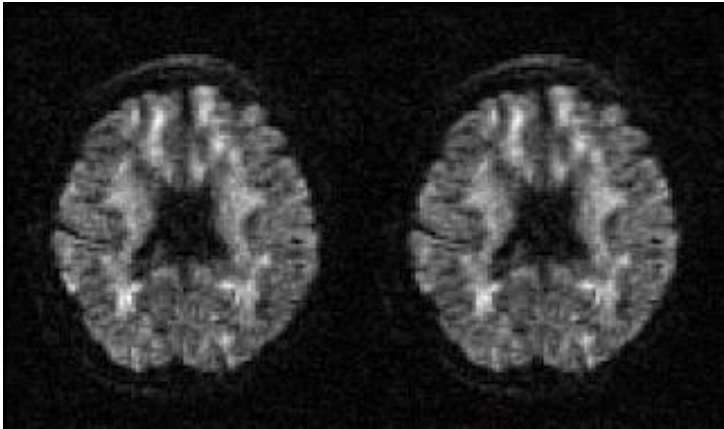
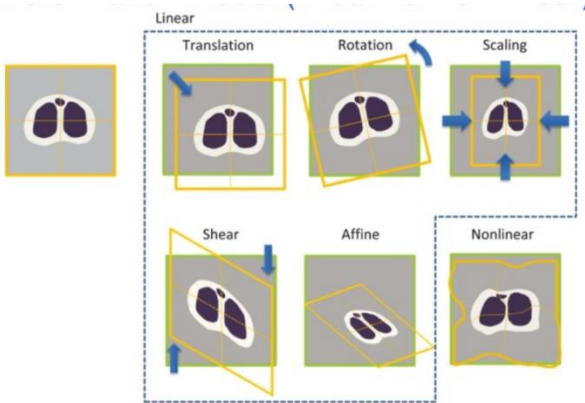
Affine Transformation

	Before	After	Preserves
Translation (3 dof)			orientation
Rotation (3 dof)			lengths
Rigid (6 dof)			lengths
Similarity (7 dof)			angles
Affine (12 dof)			parallelism

Homogeneous Transformations

Serge Koudoro

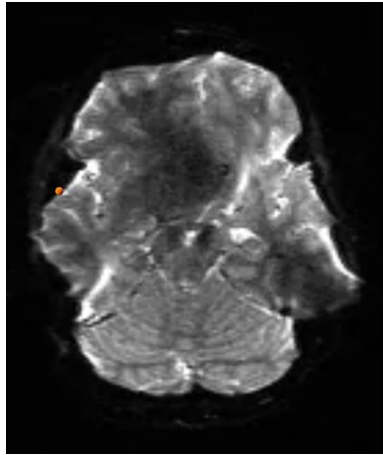
DIPY



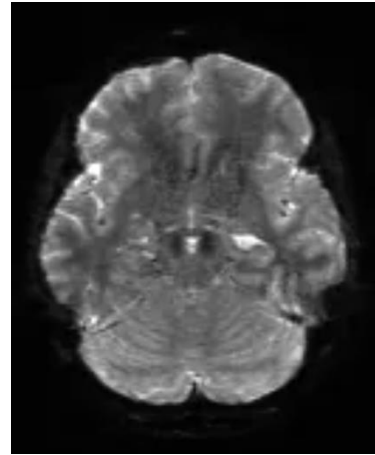
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Brain Distortion



dMRI
>>>



23.1× faster than topup.

Wall-clock to estimate the field on identical input.



<https://sreekar.ch/sucor/>



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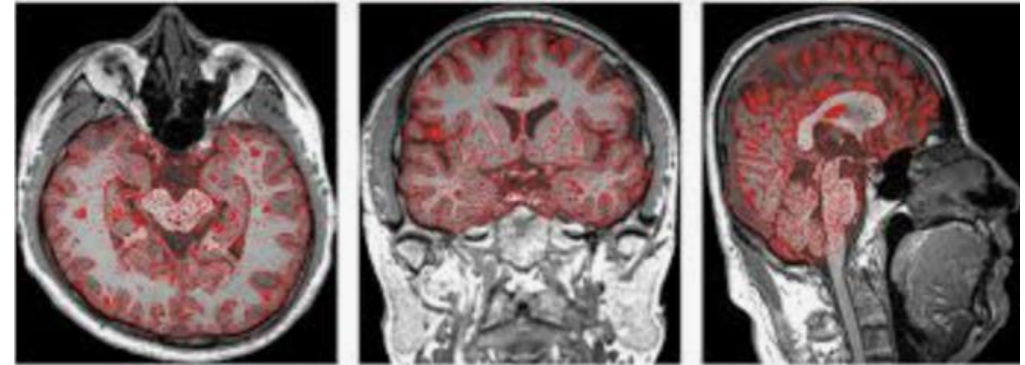
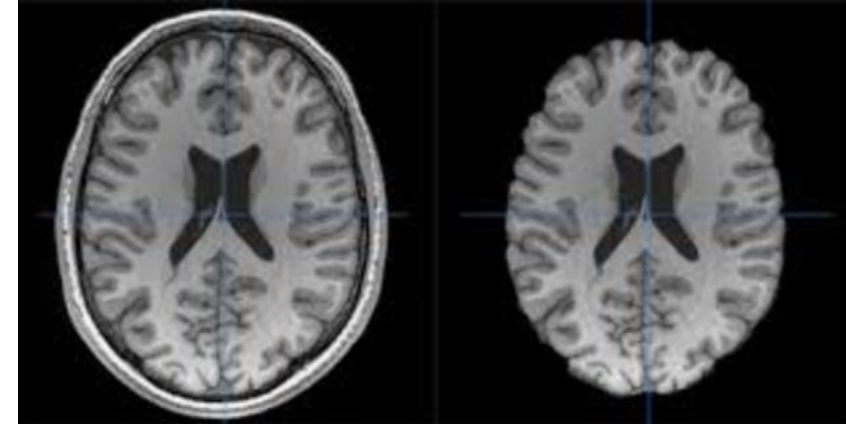
Brain extraction

```
>>> Dipy_brain_mask
```

Method:

- median_otsu (for dwi)
- evac_plus (for T1)
- SynthSeg

BET



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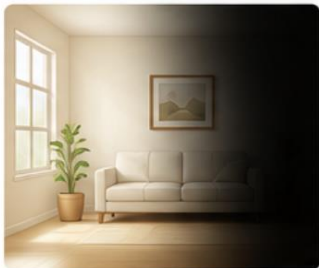
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Biasfield Correction

THE PROBLEM

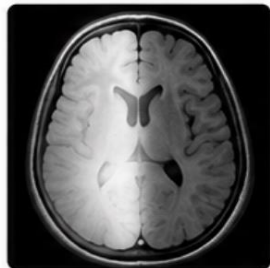
MRI images often suffer from **intensity inhomogeneity** (bias field):
same tissue appears brighter in some areas and darker in others.

Real-world analogy



Uneven lighting in a room

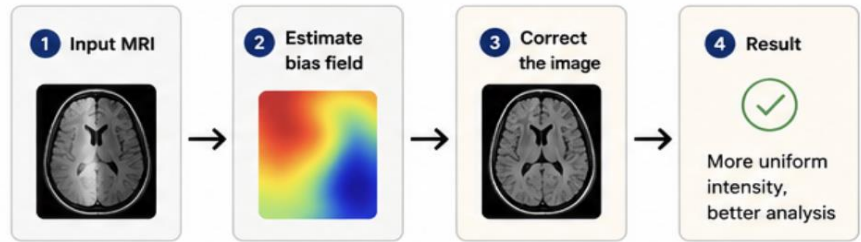
MRI analogy



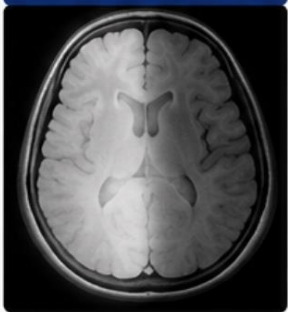
Same tissue, different intensity

WHAT IS BIAS FIELD CORRECTION?

The bias field is a smooth, slowly varying effect.
We estimate this smooth field and remove it.

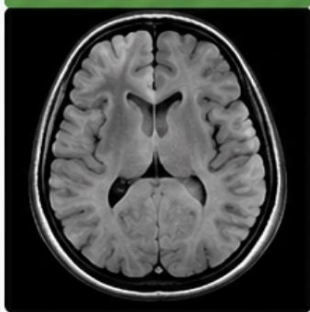


1. BIASED MRI (INPUT)



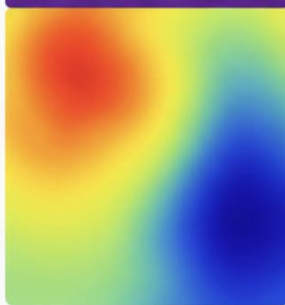
=

2. TRUE IMAGE (UNKNOWN)



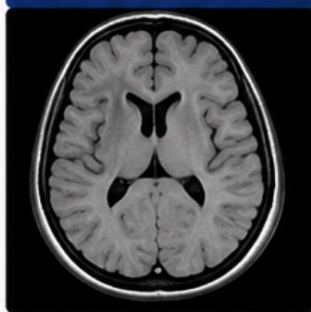
×

3. BIAS FIELD (SMOOTH)



÷

4. CORRECTED MRI (OUTPUT)



GOAL

Remove the bias field so the same tissue has similar intensity everywhere.

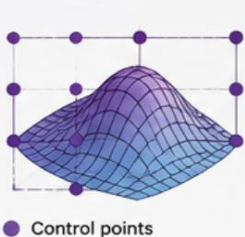
This improves:

- ✓ Segmentation
- ✓ Quantification
- ✓ Comparison across subjects / time
- ✓ Machine learning models

HOW N4 WORKS (SIMPLIFIED)

- 1 Take the log of the image.
- 2 Model the bias field as a smooth function using B-splines.
- 3 Iteratively estimate the bias field by minimizing high-frequency content in the corrected image.
- 4 Divide the original image by the estimated bias field.
- 5 Get a bias-corrected image with more uniform intensity.

B-spline idea

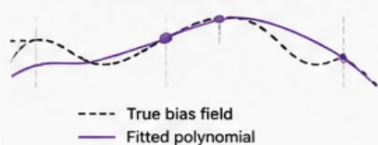


The bias field is represented by a smooth surface controlled by a grid of points.
Moving one point affects only the nearby region.

● Control points

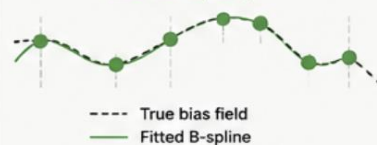
SPLINE vs POLYNOMIAL

POLYNOMIAL (global)



- ✗ One global equation
- ✗ Changes in one place affect the whole image
- ✗ Less flexible for complex bias fields

B-SPLINE (local)



- ✓ Many local control points
- ✓ Local changes affect only nearby areas
- ✓ More flexible and accurate

WHY DOES BIAS FIELD HAPPEN?



RF coil sensitivity variations



Magnetic field imperfections



Patient anatomy and properties



Scanner and acquisition factors

>>> Dipy_correct_biasfield



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Huge topic

Register T1- Template

```
>>> dipy_align_affine  
>>> dipy_align_syn
```

Tax CMW, Bastiani M, Veraart J, Garyfallidis E, Okan Irfanoglu M. What's new and what's next in diffusion MRI preprocessing. Neuroimage. 2022 Apr 1;249:118830. doi: 10.1016/j.neuroimage.2021.118830. Epub 2021 Dec 26. PMID: 34965454; PMCID: PMC9379864.

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9379864/>

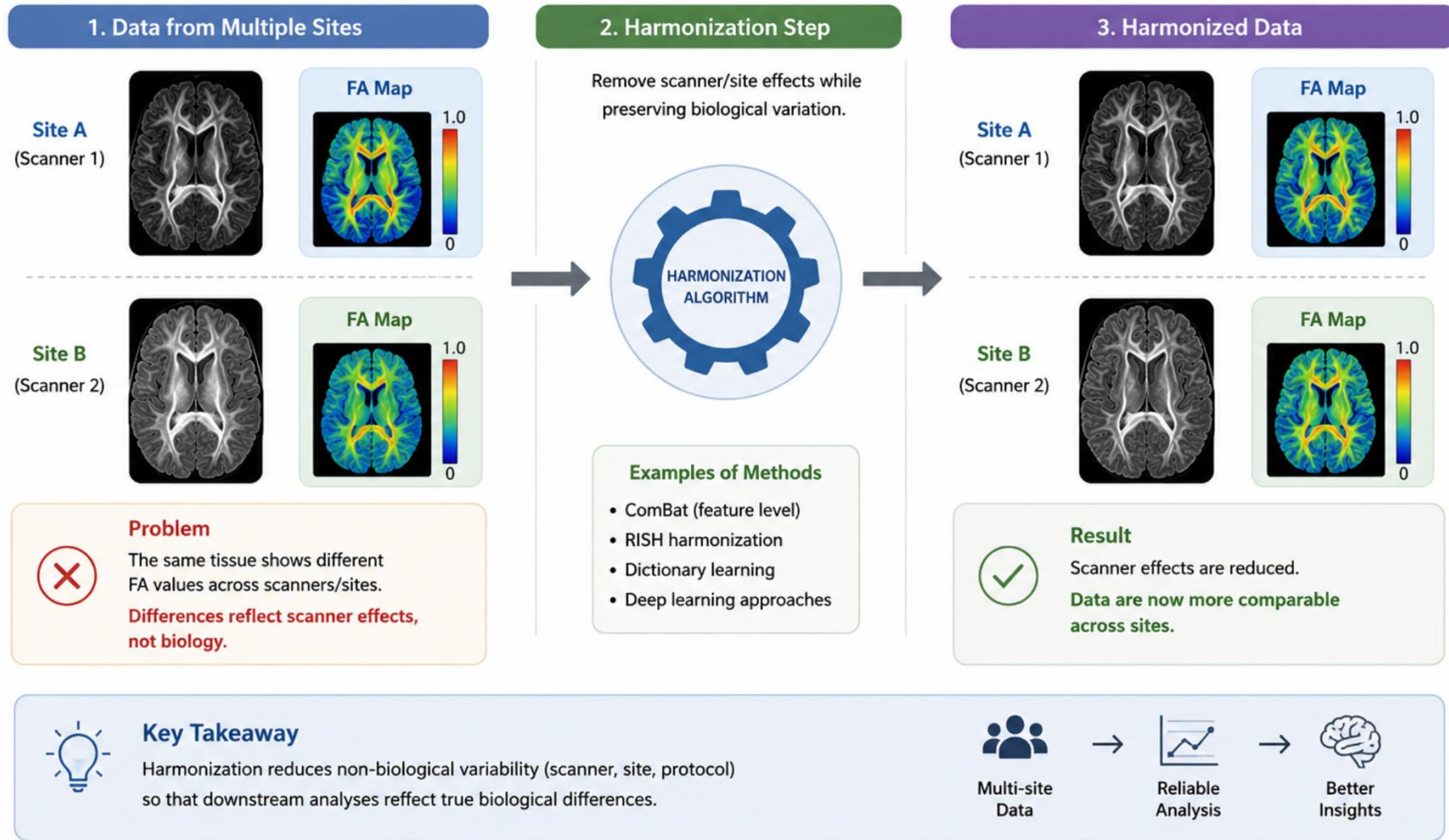


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Data Harmonization in Diffusion MRI

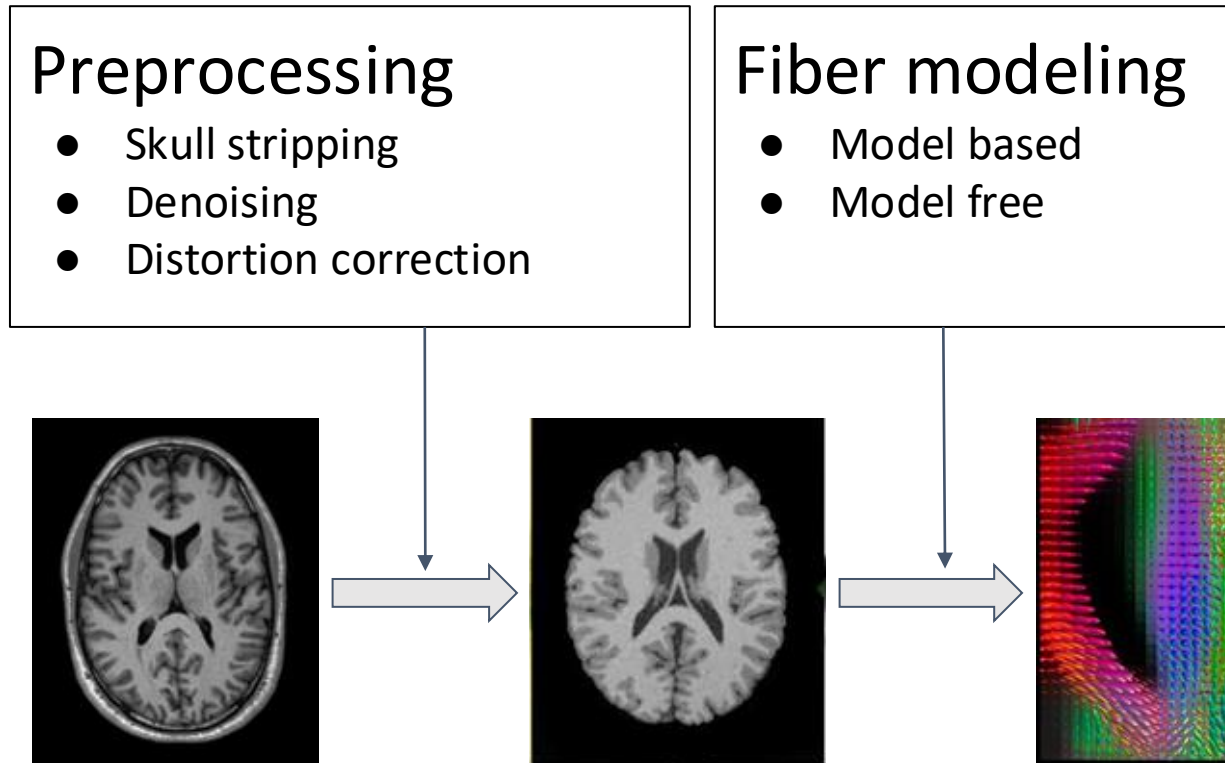
Goal: remove scanner/site differences so data become comparable



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dMRI pipeline: Reconstruction



➔ Reconstruction

```
>>> dipy_fit_dti
>>> dipy_fit_csa
>>> dipy_fit_csd
>>> dipy_fit_dki
>>> dipy_fit_dsi
>>> dipy_fit_ivim
>>> dipy_fit_mapmri
>>> dipy_fit_forecast
>>> dipy_fit_rumba
>>> ....
>>> 30+ models
```



A bit of theory

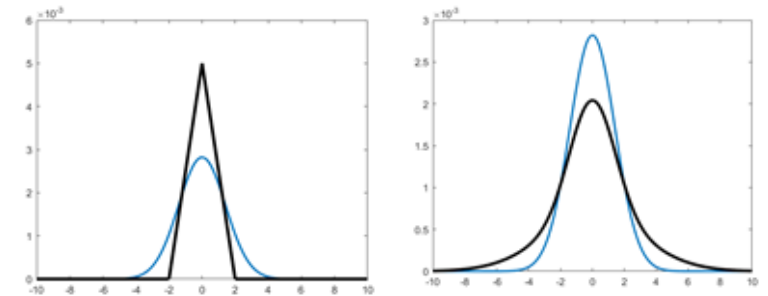
- **Phenomenological models** - Represent observable properties without assuming hidden mechanisms
 - eg - DTI, DKI
- **Mechanistic models** - Postulate hidden mechanism. In case of brain, fiber orientations and density of neural tissues
 - eg - NODDI

Gaussian diffusion Modeling

DTI $\ln \left(\frac{S(\theta)}{S_0} \right) = -b \mathbf{g}^T \mathbf{D} \mathbf{g}$

DKI $\log \frac{S(b,n)}{S_0} \approx -bD(n) + \frac{1}{6}b^2D(n)^2K(n)$

DTI has 6 parameters
DKI has 15 parameters

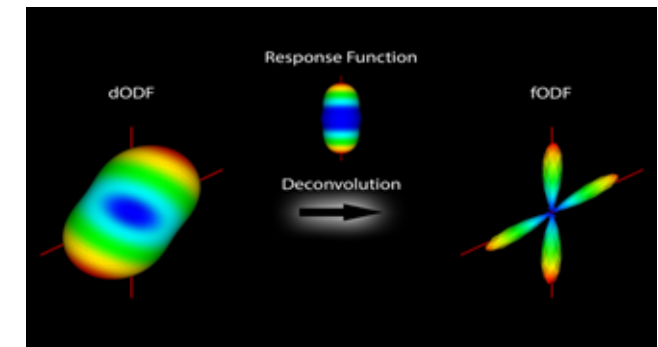


Orientation Distribution Modeling

CSA $ODF(\mathbf{n}) = \int_{\mathbb{R}^3} E(\mathbf{q}) \delta(\mathbf{q} \cdot \mathbf{n} - r) d\mathbf{q}$

CSD $S(\mathbf{q}) = \int_{S^2} R(\mathbf{q}, \mathbf{n}) f(\mathbf{n}) d\mathbf{n}$

...

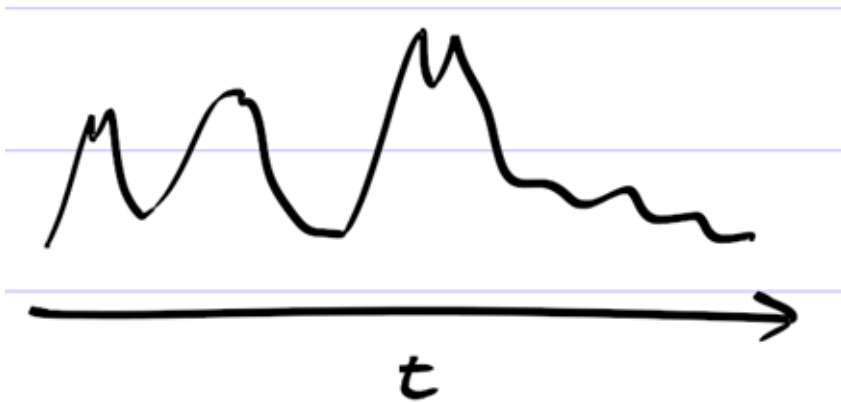


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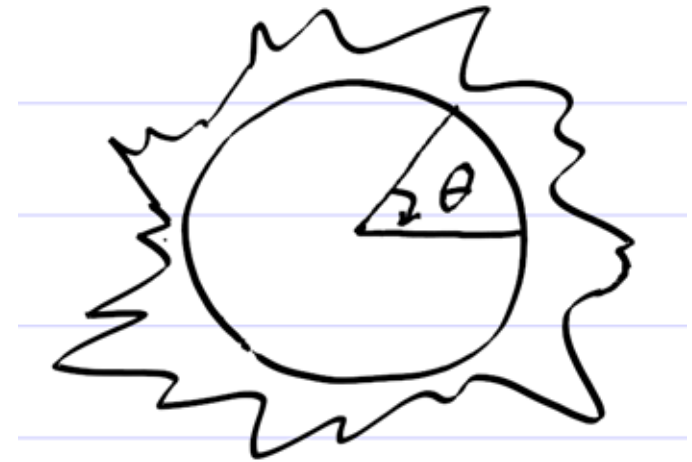
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Demystifying Spherical Harmonics

- ❑ A spherical harmonic transform is simply the Fourier Transform constrained by the fact that the domain of the signal is a sphere.
- ❑ Spherical harmonics are Fourier basis functions on the sphere.



Standard 1D signal where we could
apply the Fourier Transform



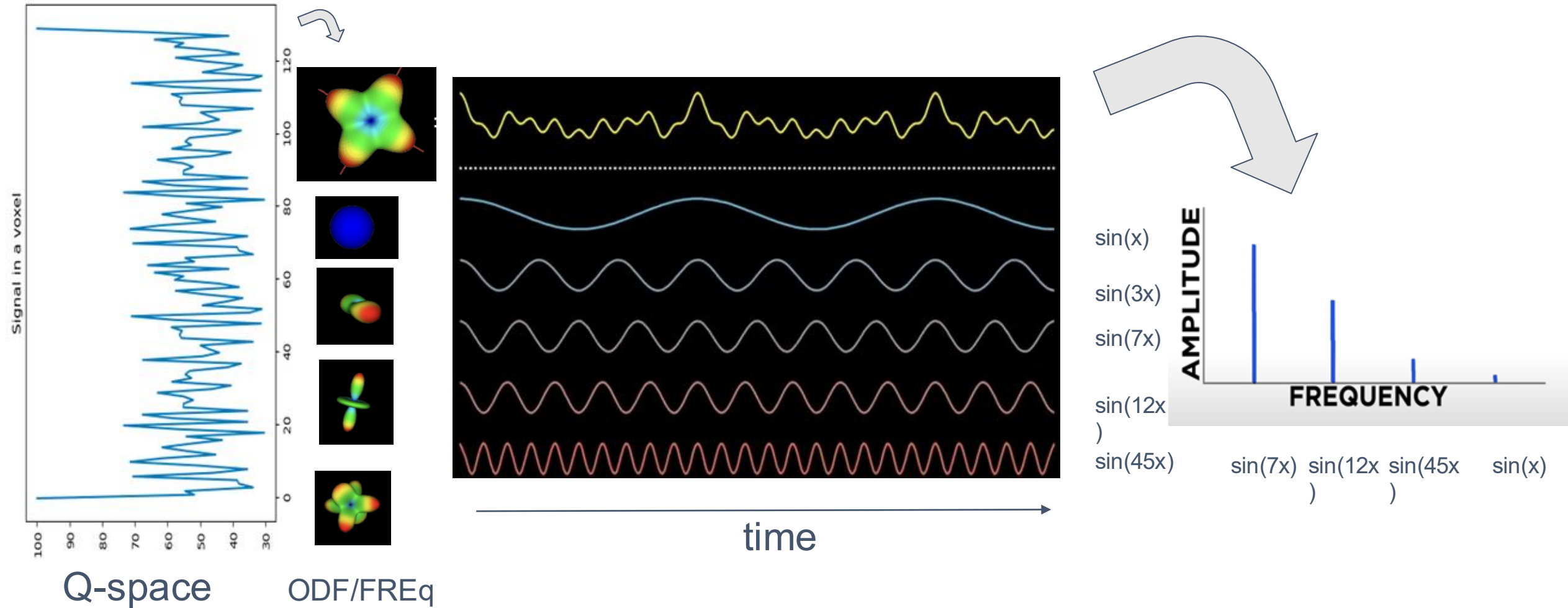
Spherical signal (showing only a slice
of that sphere) - Signal wraps around
the sphere.



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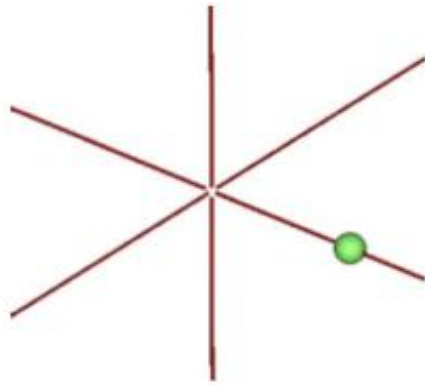
But what is the Fourier Transform? $\mathcal{F}\{f(t)\} = F(\xi) = \int_{-\infty}^{\infty} f(t) e^{-2\pi i \xi t} dt$



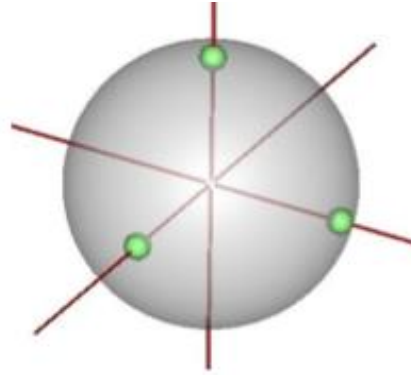
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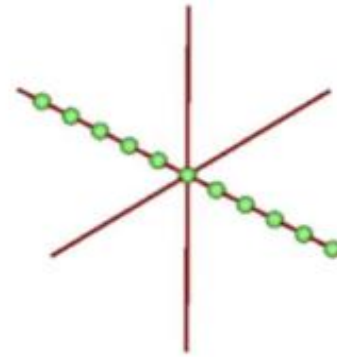
History of q-space sampling



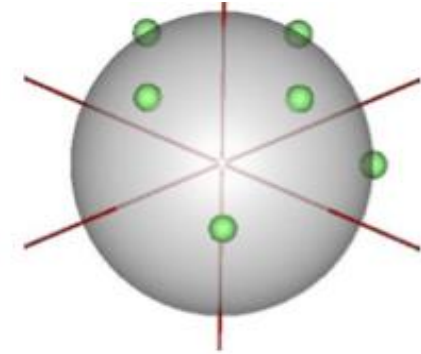
Pulsed Gradient Spin Echo
Stejskal & Tanner, 1965



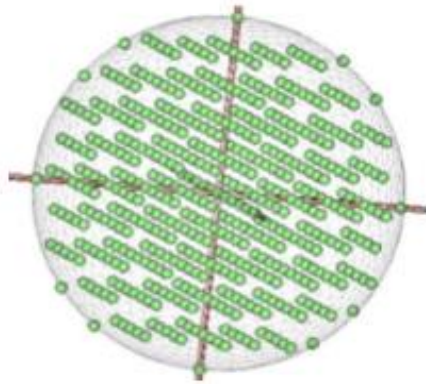
Diffusion-weighted imaging
Le Bihan, 1986



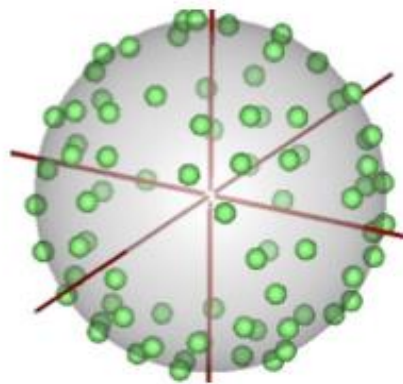
Diffusion spectrum
Callaghan, 1991



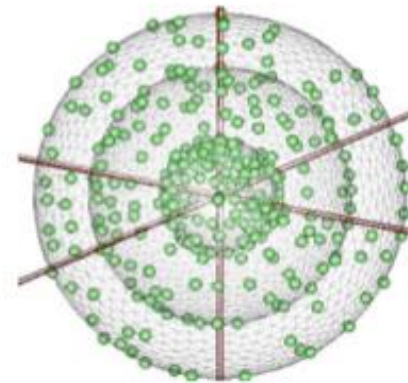
Diffusion tensor imaging
Basser, 1994



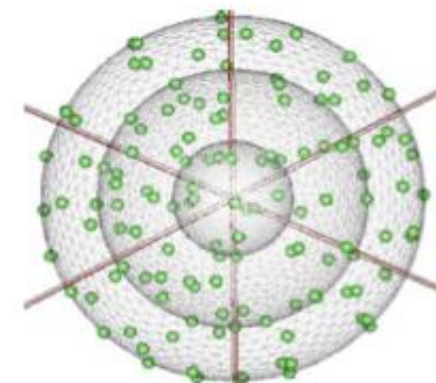
Diffusion spectrum imaging
Van Wooten, 2000



Single-Shell High Angular
Resolution Diffusion Imaging
2000-2008



Multiple-Shell Diffusion
Propagator Imaging
2008-now



Multiple-Shell, sparse
Hybrid Diffusion Imaging
2008-now

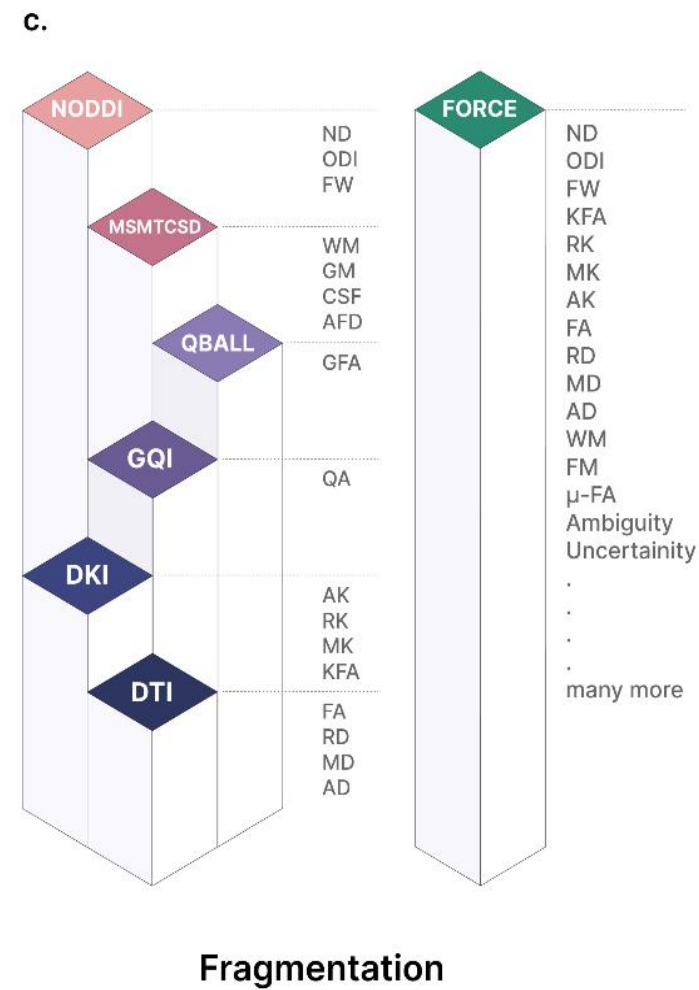
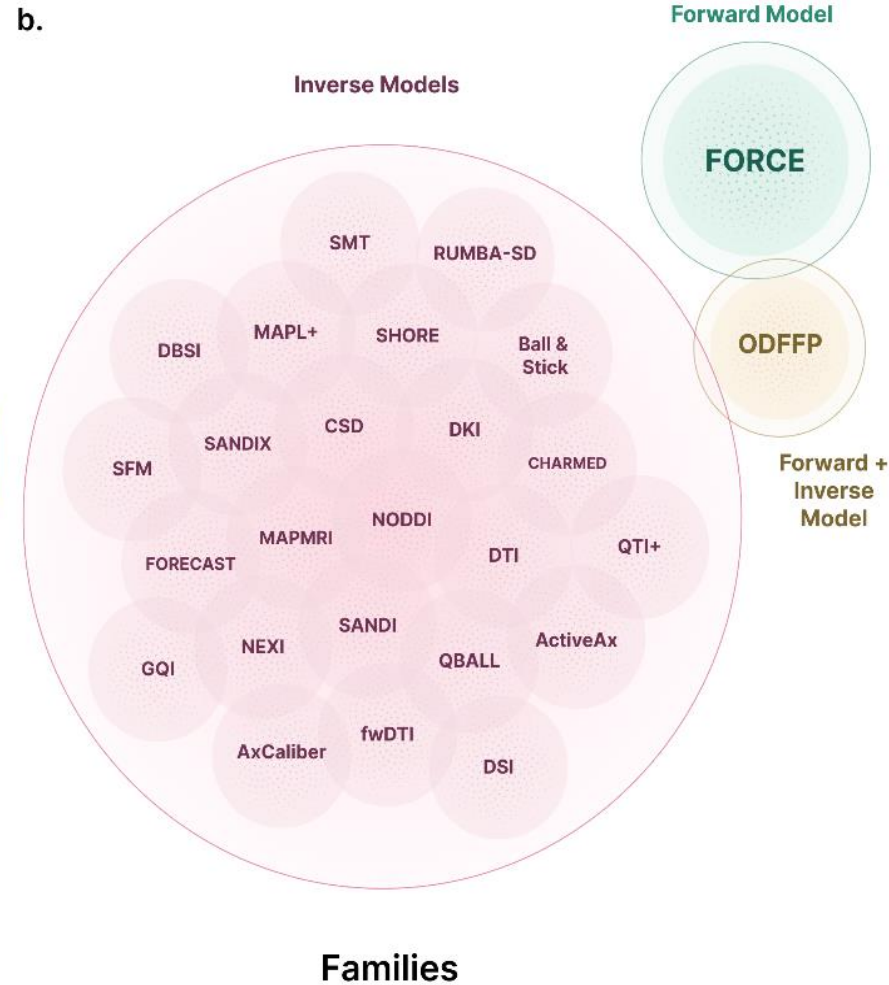
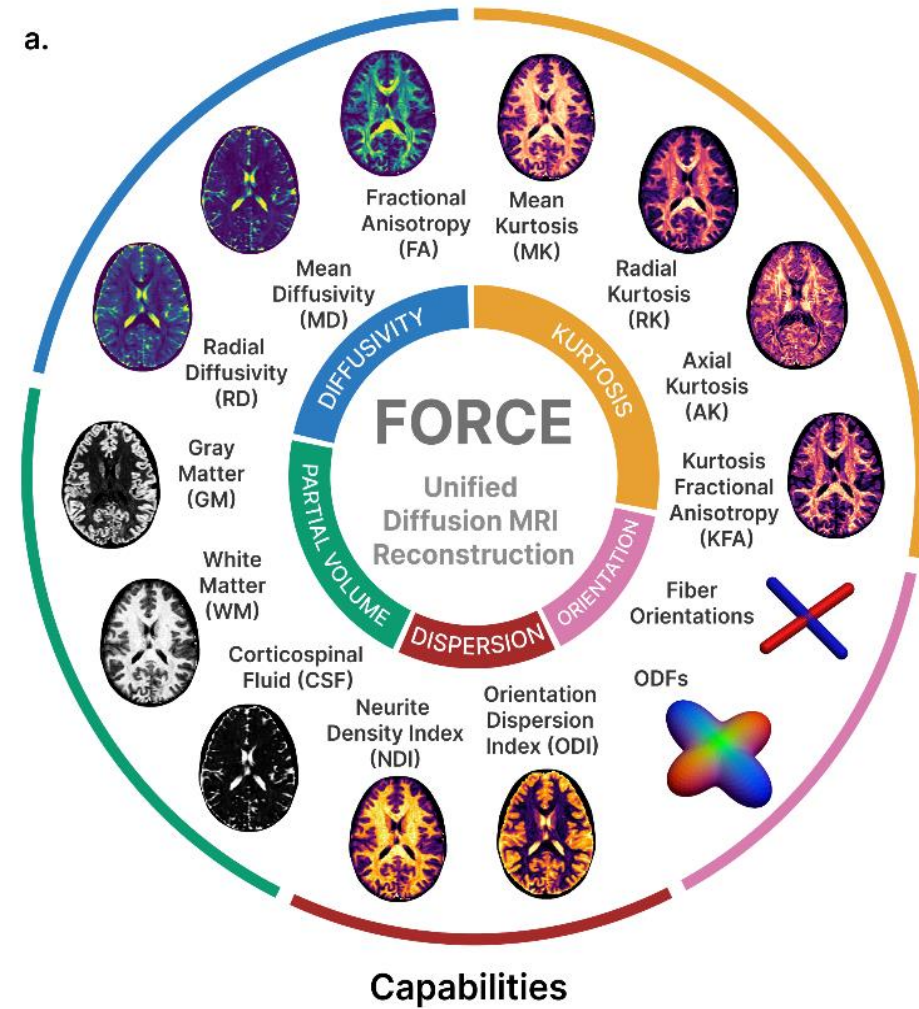
```
>>> dipy_fit_dti
>>> dipy_fit_csa
>>> dipy_fit_csd
>>> dipy_fit_dki
>>> dipy_fit_dsi
>>> dipy_fit_ivim
>>> dipy_fit_mapmri
>>> dipy_fit_forecast
>>> dipy_fit_gqi
>>> ....
>>> 30+
```

Method	Single Shell	Multi Shell	Cartesian	Paper Data Descriptions	References
DTI (SLS, WLS, NNLS)	Yes	Yes	Yes	Typical b-value = 1000 s/mm ² , maximum b-value 1200 s/mm ² (some success up to 1500 s/mm ²)	Basser <i>et al.</i> [BMLeBihan94b]
DTI (RESTORE)	Yes	Yes	Yes	Typical b-value = 1000 s/mm ² , maximum b-value 1200 s/mm ² (some success up to 1500 s/mm ²)	Chang <i>et al.</i> [CJP05], Chung <i>et al.</i> [CLH06], Yendiki <i>et al.</i> [YKK+14]
FwDTI	No	Yes	No	DTI-style acquisition, multiple b=0, all shells should be within maximum b-value of 1000 s/mm ² (or 32 directions evenly distributed 500 s/mm ² and 1500 s/mm ² per Neto Henriques <i>et al.</i> [NetoHenriquesRG+17])	Pasternak <i>et al.</i> [PSG+09], Neto Henriques <i>et al.</i> [NetoHenriquesRG+17]
DKI - Standard	No	Yes	No	Dual spin echo diffusion-weighted 2D EPI images were acquired with b values of 0, 500, 1000, 1500, 2000, and 2500 s/mm ² (max b value of 2000 suggested as sufficient in brain tissue); at least 15 directions	Jensen <i>et al.</i> [JHR+05]
DKI+ Constraints	No	Yes	No	None	Dela Haije <i>et al.</i> [DelaHaijeOzarslanF20]
DKI - Micro (WMTI)	No	Yes	No	DKI-style acquisition: at least two non-zero b shells (max b value	Fieremans <i>et al.</i> [FJH11], Tabesh <i>et al.</i> [TJAH11]

↑ Back to top

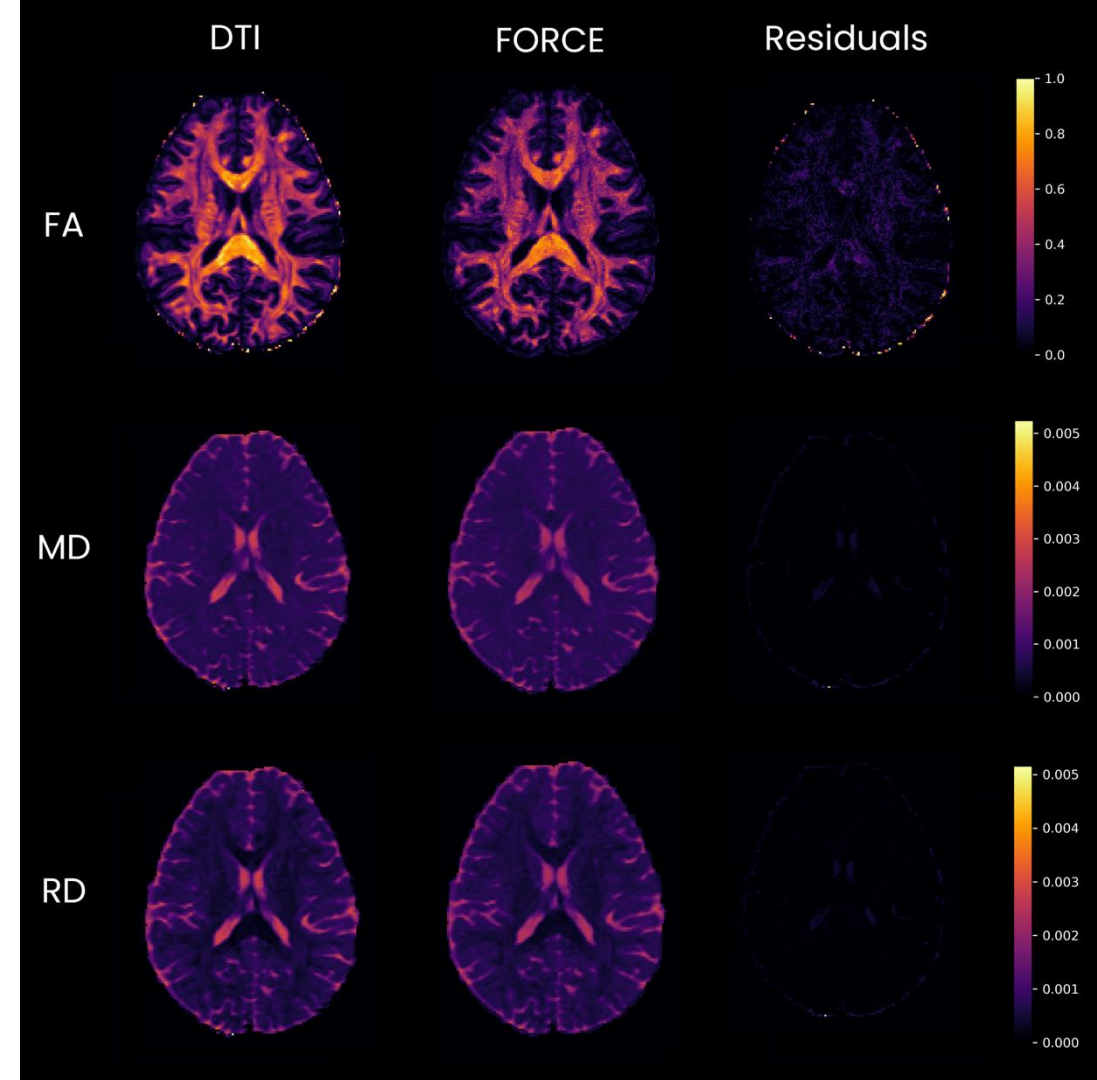
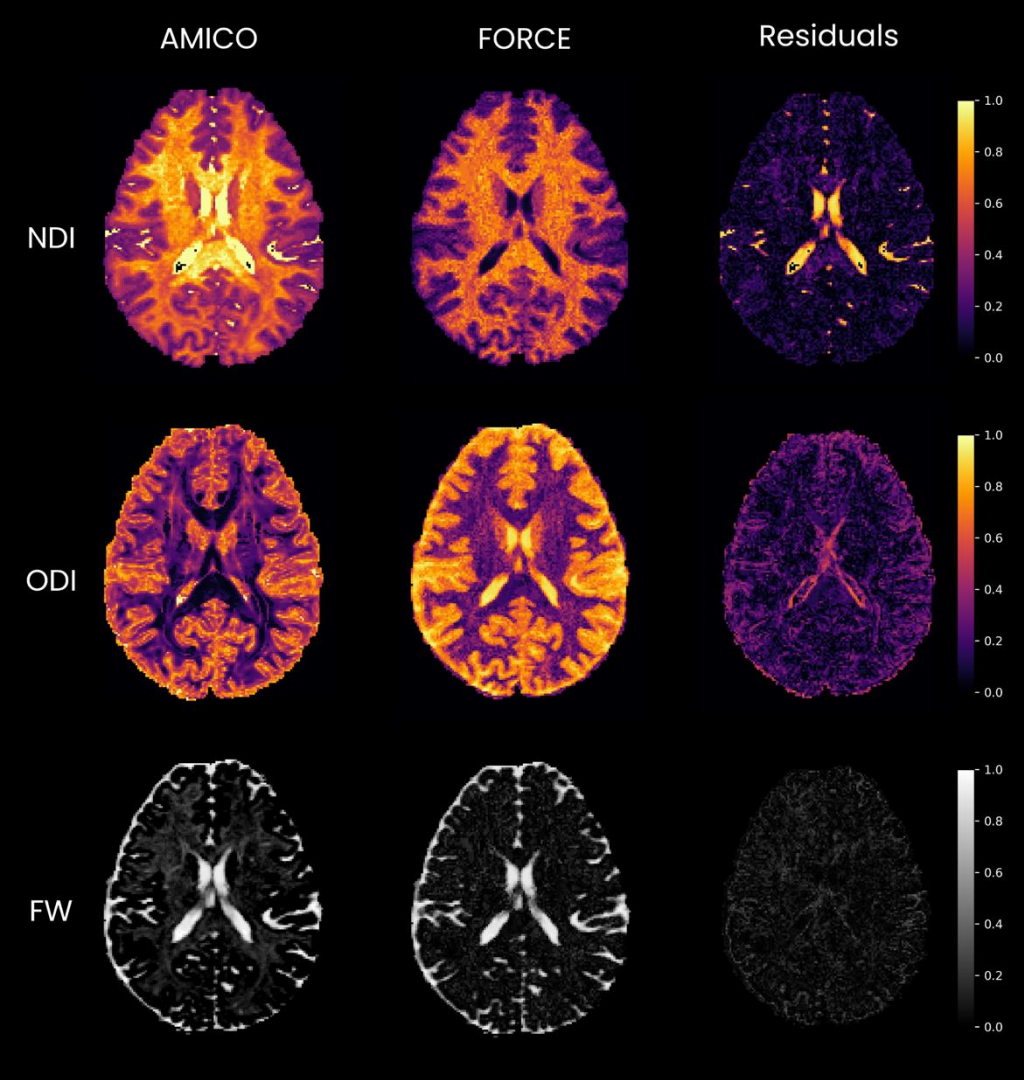


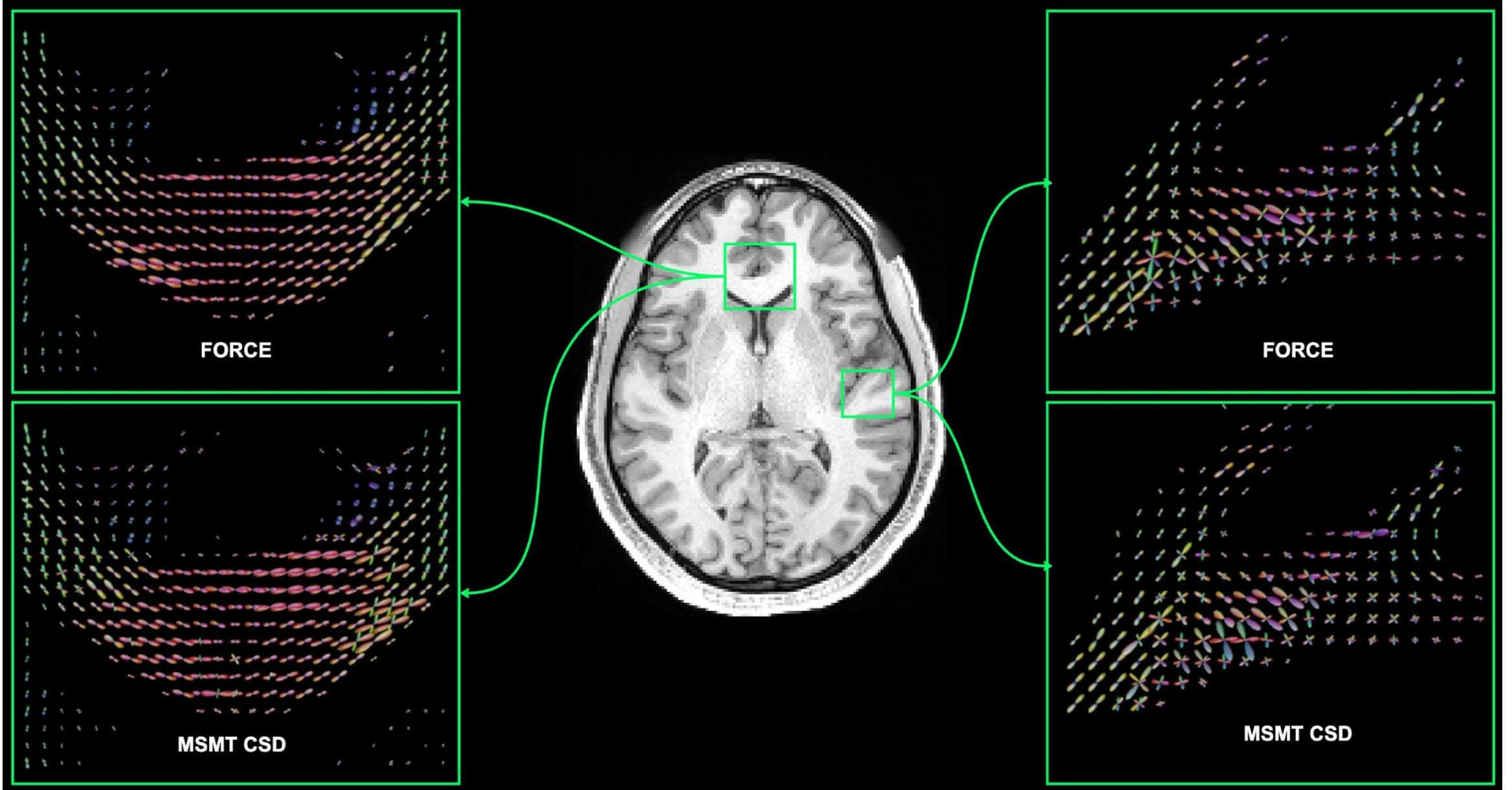
> DIPY_fit_force

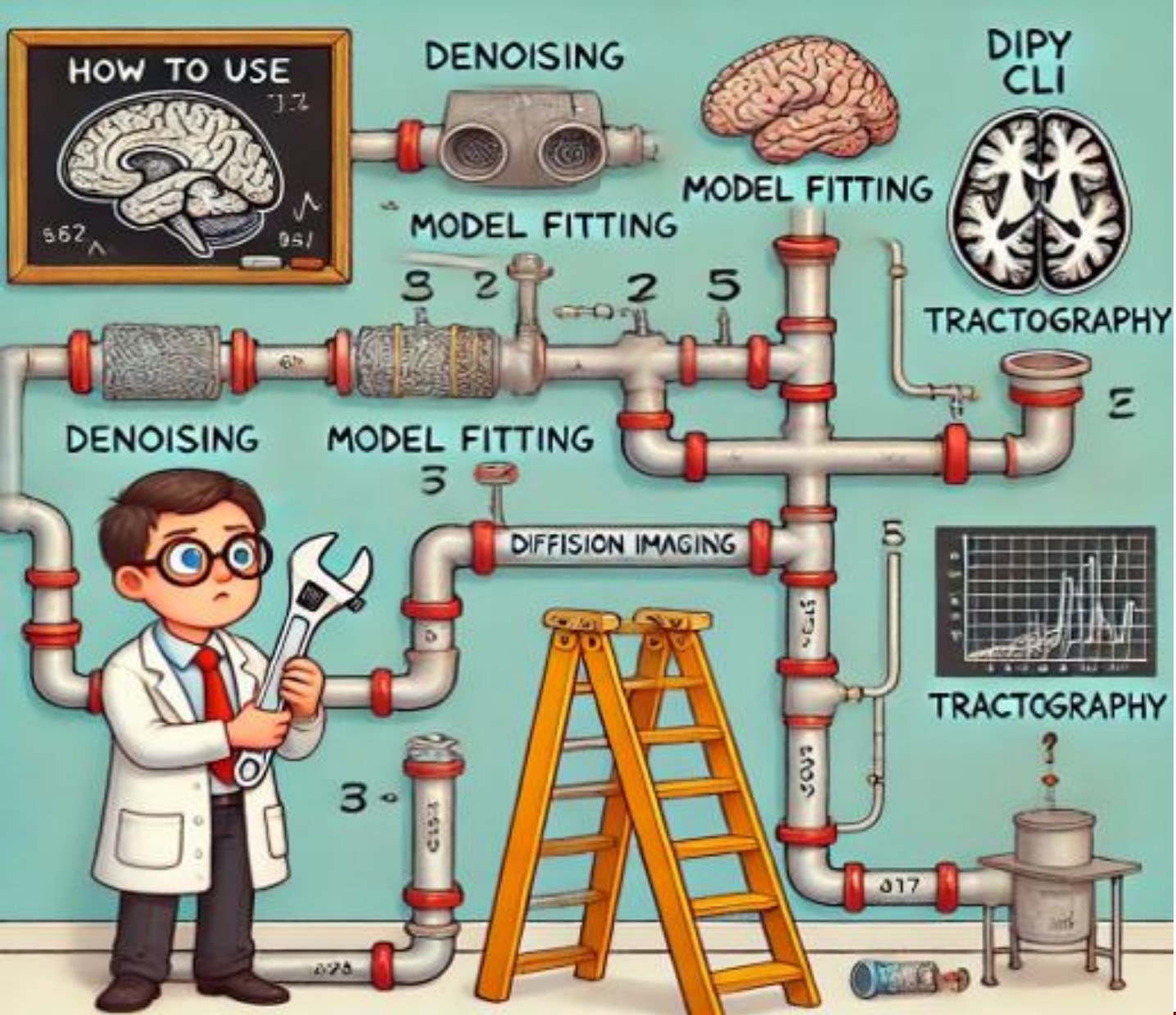


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How to build your diffusion imaging pipeline



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Introducing DIPY AUTO



Yet another pipeline ?



QSIPREP



QSIRECON



TRACTOFLOW



MRTRIX



SF-PEDIATRICS



TRACULA



TORTOISE



SCILPY



FSL



& MORE.....



Reality



Different Data "Quality" ?



The "Anatomy" Problem



Complexity to adapt pipelines



Rule of thumb

there is no "one size fits all" solution



What drives DIPY-AUTO?

1

Community Request

2

Needs of flexibility

3

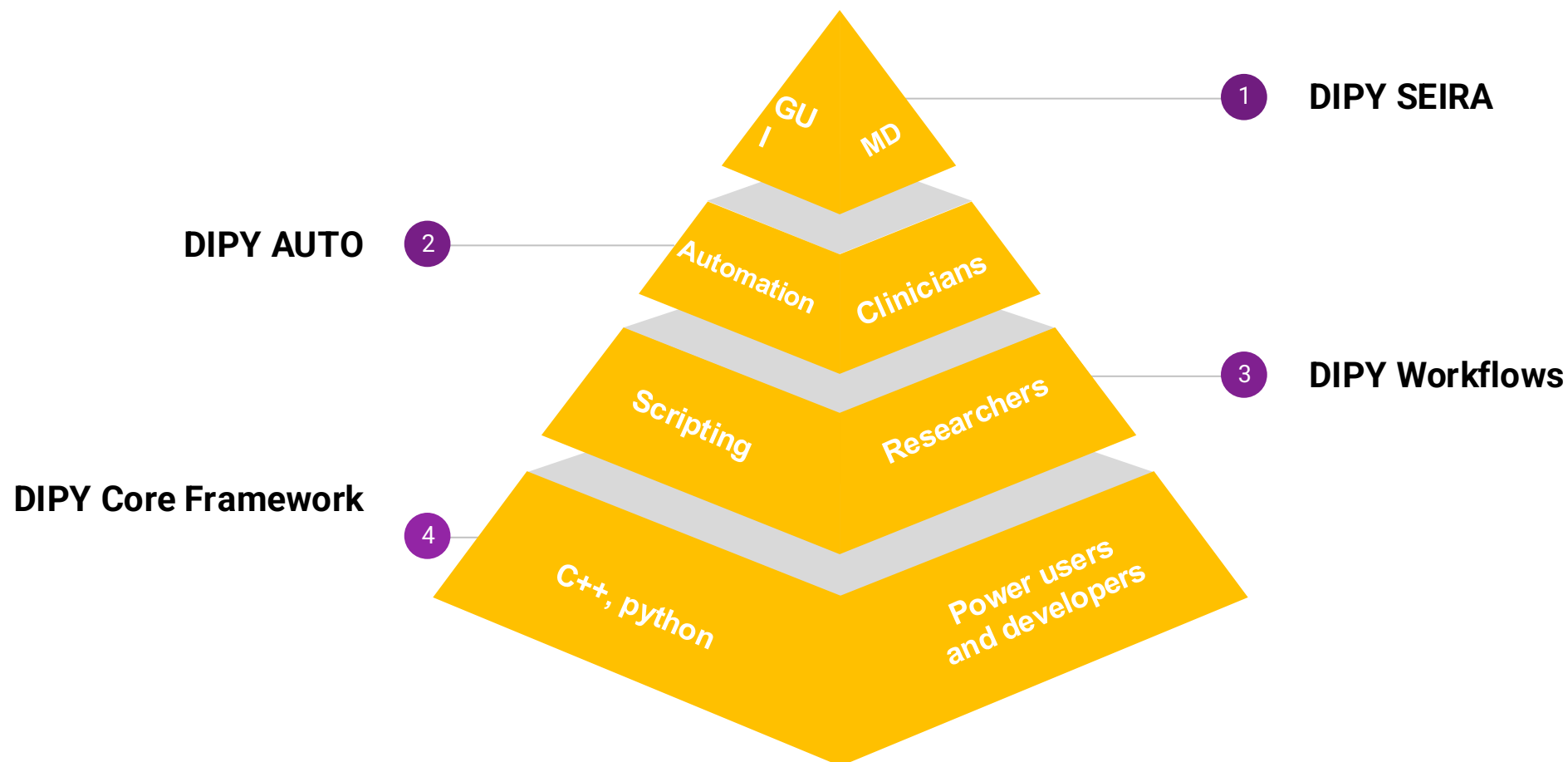
Needs of accessibility



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How is it Built ?



How DIPY AUTO works ?



Predefined Mode

POPULAR

Best for

Standard clinical pipelines and beginners



Recommended by DIPY Core Team



Not Flexible



Interactive Mode

EXPERIMENT

Best for

Quick prototyping and parameter tuning



Experiment Fast



Limited access to specific methods.



Custom Mode

FLEXIBLE

Best for

Power users and novel research applications



Maximum Flexibility.



Requires expertise; higher risk of errors.

 To run it : `> dipy_auto dwi.nii.gz dwi.bval dwi.bvec -out_dir result_dwi`





To run it : > `dipy_auto dwi.nii.gz dwi.bval dwi.bvec -interactive -out_dir result_dwi`

Pipeline Selection

Your data has 4 unique b-values: [0.0, 1000.0, 2000.0, 3500.0]

Recommended pipeline: 'multi_shell'

Recommended methods: dti, dki, csd, csa, gqi

Available predefined pipelines:

- 0. Custom (interactive) - Build your own pipeline step-by-step
- 1. full - Full pipeline: preprocessing + reconstruction + tracking + SLR + bundles

Enter choice [0-1] (or press Enter for 'full'): 0

[2026-03-16 04:48:08][dipy] INFO: Building custom pipeline interactively...

Interactive Pipeline Builder

Your data has 4 unique b-values: [0.0, 1000.0, 2000.0, 3500.0]

Recommended pipeline: 'multi_shell'

Recommended methods: dti, dki, csd, csa, gqi

Preprocessing Steps Selection

Select preprocessing steps to include (Brain Mask Extraction always runs):

- 1. Denoising (NLMeans)
- 2. Denoising (MPPCA)
- 3. Denoising (LPCA)
- 4. Denoising (Patch2Self)
- 5. Reslice to isotropic voxels
- 6. Gibbs ringing removal
- 7. Motion correction (eddy)
- 8. Bias field correction

Reconstruction Methods Selection

[2026-03-16 04:49:11][dipy] INFO:

Select reconstruction methods (DTI always runs):

- [2026-03-16 04:49:11][dipy] INFO: 1. FORCE - FORward modeling for Complex microstructure Estimation (✓ compatible)
- [2026-03-16 04:49:11][dipy] INFO: 2. DKI - Diffusion Kurtosis Imaging (✓ compatible)
- [2026-03-16 04:49:11][dipy] INFO: 3. CSD - Constrained Spherical Deconvolution (✓ compatible)
- [2026-03-16 04:49:11][dipy] INFO: 4. CSA - Constant Solid Angle (✓ compatible)
- [2026-03-16 04:49:11][dipy] INFO: 5. GQI - Generalized Q-Sampling Imaging (✓ compatible)
- [2026-03-16 04:49:11][dipy] INFO: 6. MAPMRI - Mean Apparent Propagator MRI (✓ compatible)

[2026-03-16 04:49:11][dipy] INFO:

Enter numbers (comma-separated, e.g., 1,2) or press Enter for all:

Methods: 1,3, 6

Fiber Tracking

Include fiber tracking? [Y/n]: Y

Choose brain extraction method:

- 1. SynthSeg - More accurate, requires pytorch installed
- 2. median_otsu - Good accuracy, no extra dependencies

Enter choice [1-2]:

Multiple reconstruction methods available for tracking:

- 1. dti
- 2. force
- 3. csd
- 4. mapmri

Enter numbers (comma-separated, e.g., 1,2) or press Enter for all:

Tracking methods:



To run it : > `dipy_auto config.toml`

```
[General]
name = "full_pipeline"
description = "Full pipeline: preprocessing + reconstruction + tracking + SLR + bundles"
version = "1.0.0"
author = "DIPY Developers"
```

```
[io]
dwi = ""
bvals = ""
bvecs = ""
mask = ""
tlw = ""
bids_folder = ""
out_dir = "."
atlas_tractogram = ""
bundle_atlas_dir = ""
ap = ""
```

```
# Full preprocessing chain
```

```
[[pipeline]]
name = "reslice"
cli = "dipy_reslice"
input_files = "${io.dwi}"
```

Section Navigation

- [dipy_align_affine](#)
- [dipy_align_syn](#)
- [dipy_apply_transform](#)
- [dipy_brain_mask](#)
- [dipy_bun_lmm](#)
- [dipy_bun_shapes](#)
- [dipy_bun_profiles](#)
- [dipy_bundlewarp](#)
- [dipy_classify_tissue](#)
- [dipy_cluster_streamlines](#)
- [dipy_correct_motion](#)
- [dipy_correct_biasfield](#)
- [dipy_concatenate_tractograms](#)
- [dipy_convert_tractogram](#)
- [dipy_convert_tensors](#)
- [dipy_convert_sh](#)
- [dipy_denoise_nmeans](#)

Command Line Utilities Reference

dipy_align_affine

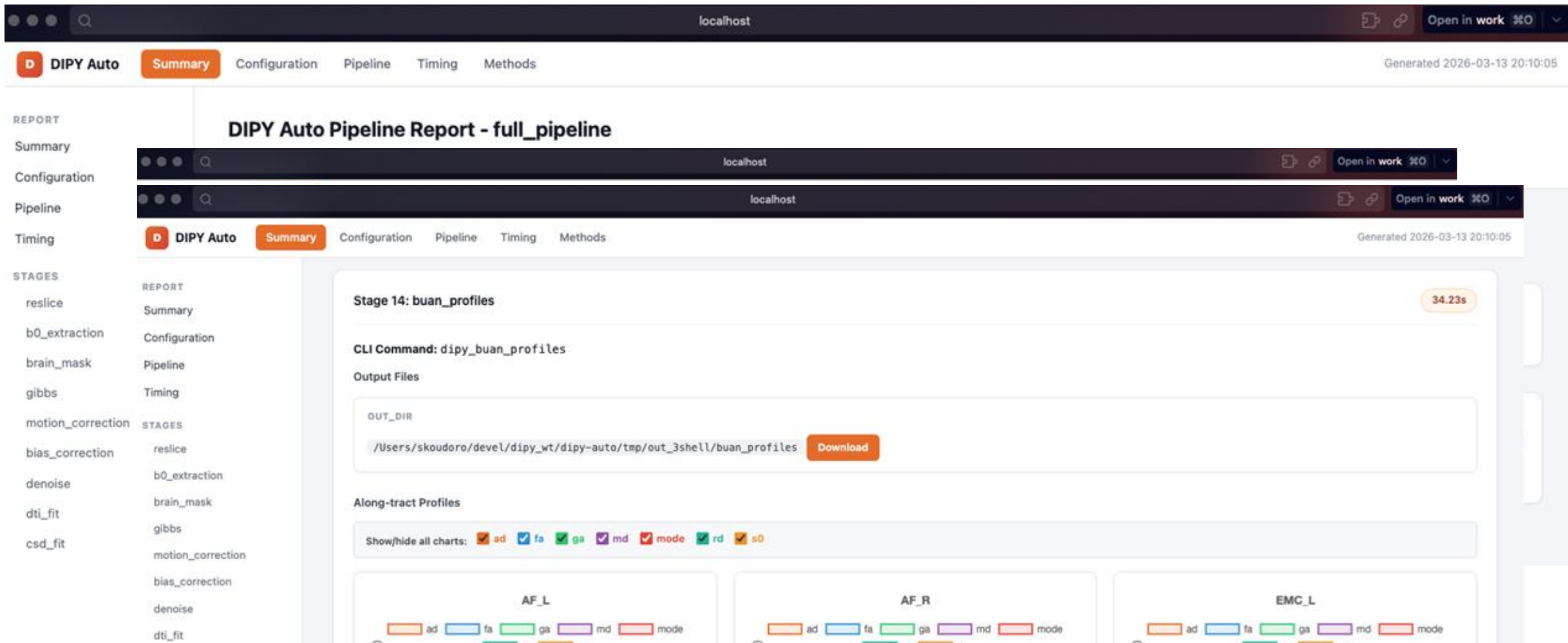
[Usage](#)[Input Parameters](#)[General Options](#)[Output Options](#)[References](#)

dipy_align_syn

[Usage](#)[Input Parameters](#)[General Options](#)[Output Options](#)[References](#)



To run it : `> dipy_auto report_20260316.html`



The screenshot displays the DIPY Auto Pipeline Report - full_pipeline interface. The top navigation bar includes tabs for Configuration, Pipeline, Timing, and Methods. The main content area shows the report details for Stage 14: buan_profiles, which took 34.23s to complete. The CLI Command is dipy_buan_profiles. The Output Files section shows the OUT_DIR: /Users/skoudoro/devel/dipy_wt/dipy-auto/tmp/out_3shell/buan_profiles, with a Download button. The Along-tract Profiles section shows a list of profiles (ad, fa, ga, md, mode) with checkboxes for Show/hide all charts. The bottom of the image shows three charts labeled AF_L, AF_R, and EMC_L, each with a legend for the profiles.

REPORT

Summary

Configuration

Pipeline

Timing

STAGES

reslice

b0_extraction

brain_mask

gibbs

motion_correction

bias_correction

denoise

dti_fit

csd_fit

DIPY Auto

Summary

Configuration

Pipeline

Timing

Methods

Generated 2026-03-13 20:10:05

DIPY Auto Pipeline Report - full_pipeline

Stage 14: buan_profiles

34.23s

CLI Command: dipy_buan_profiles

Output Files

OUT_DIR

/Users/skoudoro/devel/dipy_wt/dipy-auto/tmp/out_3shell/buan_profiles

Download

Along-tract Profiles

Show/hide all charts: ☒ ad ☒ fa ☒ ga ☒ md ☒ mode ☒ rd ☒ s0

AF_L

AF_R

EMC_L

ad fa ga md mode

Coming Next: DIPY SEIRA



Sign in to DIPY Workbench

Username or email address and password

Sign in



Select Your Data

Select DWI file (.nii.gz)

/Users/skoudoro/data/stanford_hardi/HARDI150.nii.gz

Browse

Select bvecs file (.bvec)

/Users/skoudoro/data/stanford_hardi/HARDI150.bvec

Browse

Select bvals file (.bval)

/Users/skoudoro/data/stanford_hardi/HARDI150.bval

Browse

Output Folder Path

Select output folder

Browse

← Back

Next →

1

Select Data Input & Output

Specify the source image for analysis and the target directory for saving all outputs. Proper input/output setup ensures reproducible and organized workflows.

2

Run/Build Pipeline

Customize your pipeline to match your needs. You can create a new configuration tailored to your analysis goals, or load an existing pipeline setup to save time and ensure consistency across datasets.

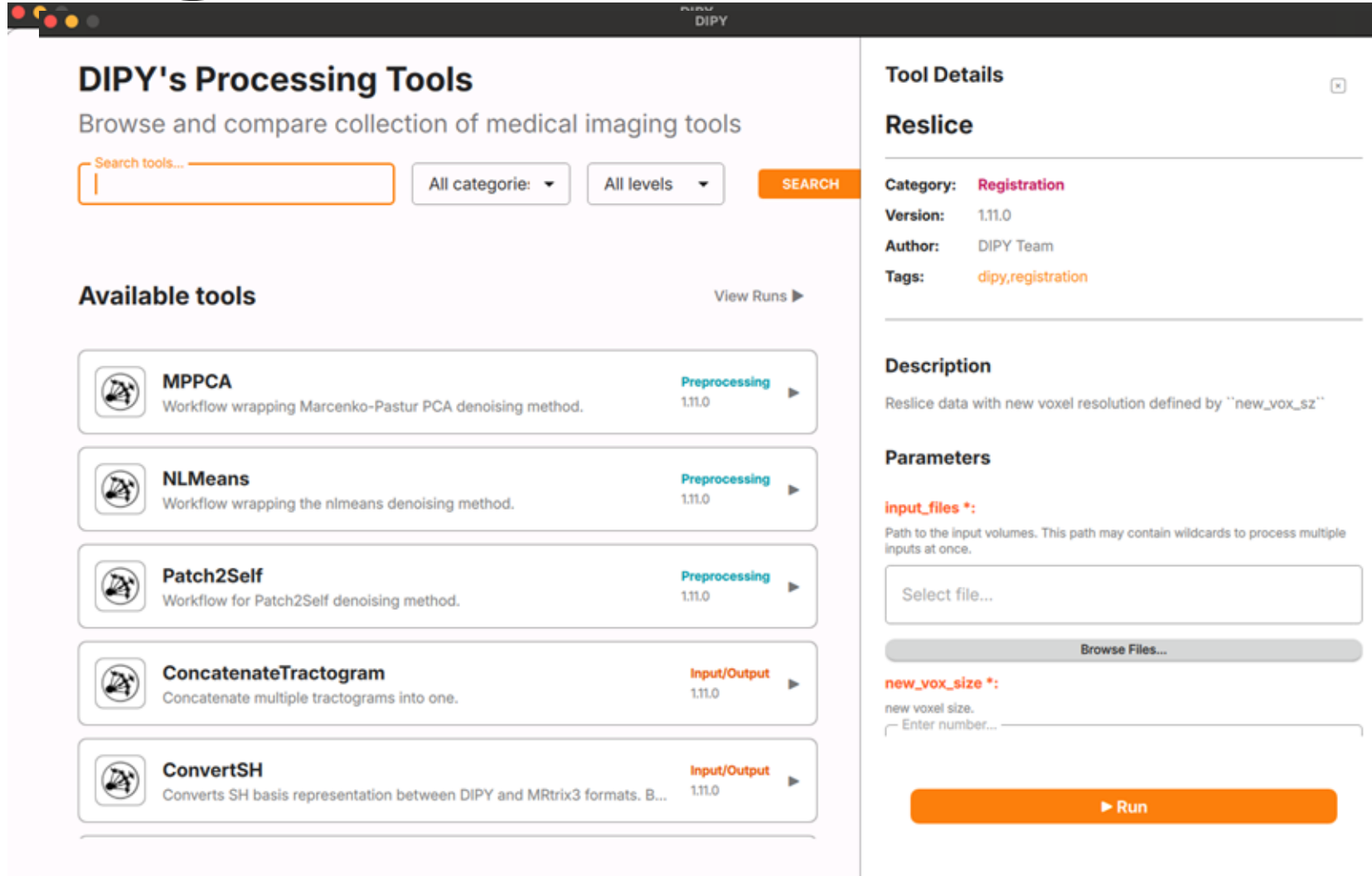
3

Explore & Analyze Results

Interact with your pipeline outputs. Visualize data, run post-processing methods, and dive deeper into results to extract insights or validate outcomes. This step supports iterative analysis and exploration.



Coming Next: DIPY SEIRA



DIPY's Processing Tools
Browse and compare collection of medical imaging tools

Search tools... All categorie: ▾ All levels ▾ **SEARCH**

Available tools [View Runs ▶](#)

-  **MPPCA** Preprocessing 1.11.0 ▶
Workflow wrapping Marcenko-Pastur PCA denoising method.
-  **NLMeans** Preprocessing 1.11.0 ▶
Workflow wrapping the nlmeans denoising method.
-  **Patch2Self** Preprocessing 1.11.0 ▶
Workflow for Patch2Self denoising method.
-  **ConcatenateTractogram** Input/Output 1.11.0 ▶
Concatenate multiple tractograms into one.
-  **ConvertSH** Input/Output 1.11.0 ▶
Converts SH basis representation between DIPY and MRtrix3 formats. B...

Tool Details

Reslice

Category: Registration
Version: 1.11.0
Author: DIPY Team
Tags: dipy,registration

Description
Reslice data with new voxel resolution defined by "new_vox_sz"

Parameters

input_files *:
Path to the input volumes. This path may contain wildcards to process multiple inputs at once.

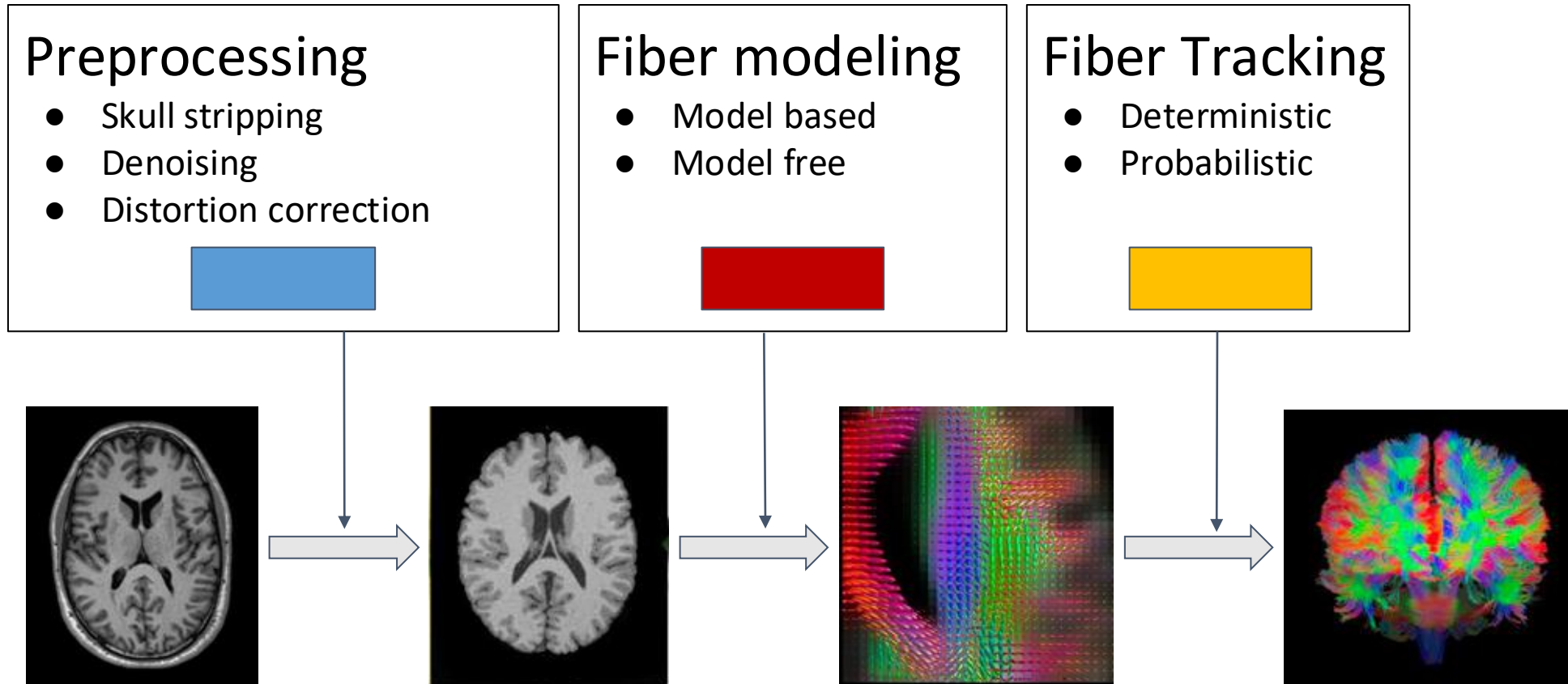
Select file...

Browse Files...

new_vox_size *:
new voxel size.
Enter number...

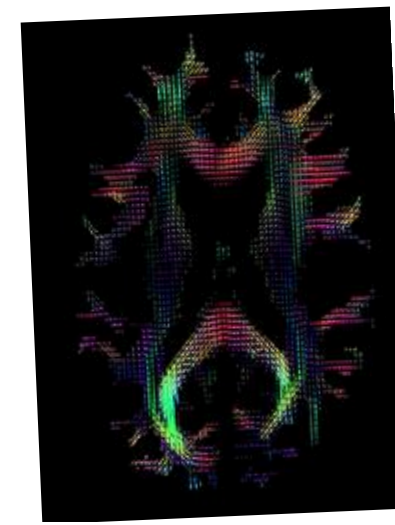
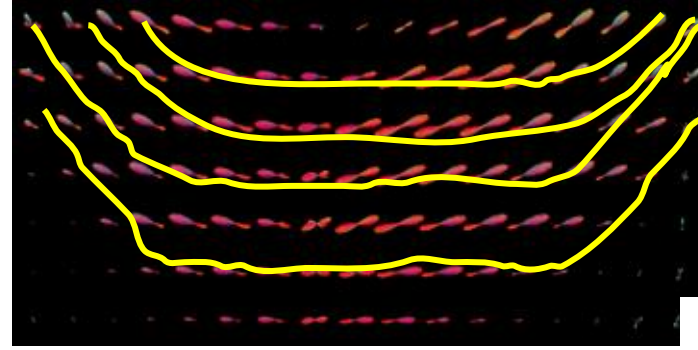
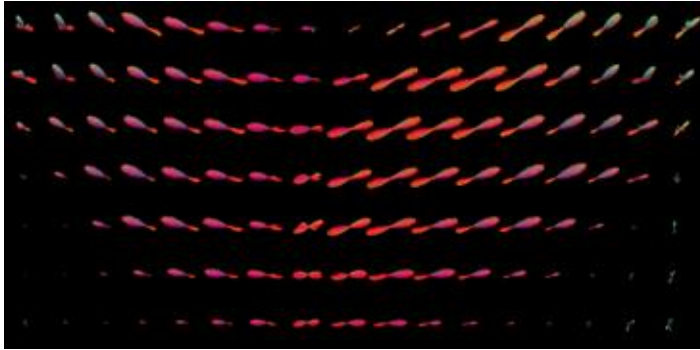
▶ Run

CLI Diffusion MRI pipeline



Tracking

A) Fiber Orientation Distribution Function B) Fiber Tracking



Requirements for streamline tractography

- Images of the white matter orientation
e.g. tensors, ODFs, fiber ODFs, peaks/maxima
- Direction selection strategy - **DirectionGetters**
e.g. highest value, probabilistic approaches
- Tracking Mask – **StoppingCriterion**
e.g. white matter volume -> estimated from the anatomical Image or DTI
- Seeding Mask
e.g. region of interest, white matter volume
- Step size, maximum curvature and interpolation method
e.g. nearest neighbour, tri-linear

	Illustration of imaging information	Unordered selection of potential tract hypotheses			Hypothetical ground truth
(A) Voxel level					
(B) Local level					
(C) Global level					

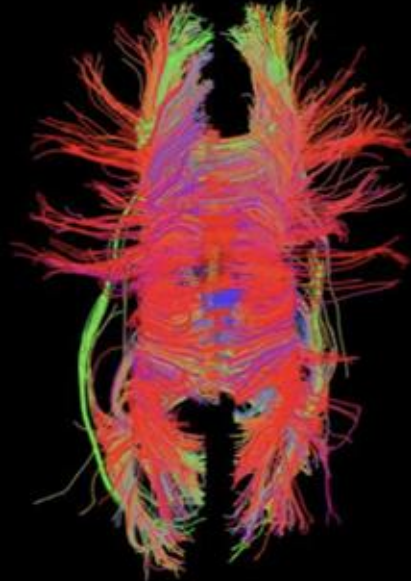
Maier-Hein, K.H., Neher, P.F., Houde, J.C. *et al.* The challenge of mapping the human connectome based on diffusion tractography. *Nat Commun* 8, 1349 (2017). <https://doi.org/10.1038/s41467-017-01285-x>



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➡ Tractography



```
--out_tractogram PFT.trk
```



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➡ Visualize Tractography results

Execution examples

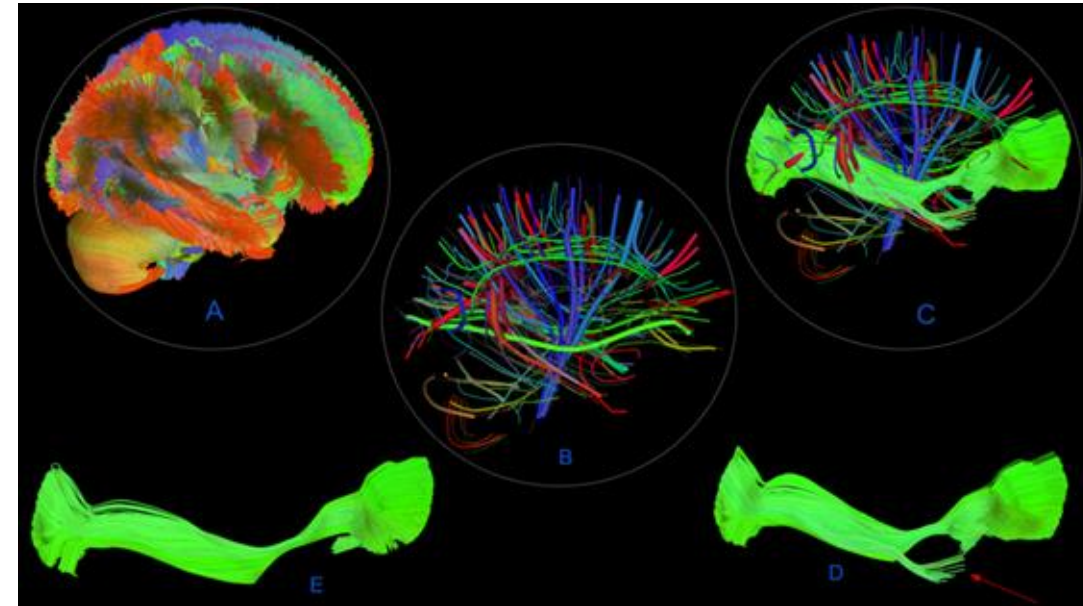
```
$ dipy_horizon tractogram.trk
```

```
$ dipy_horizon bundle1.trk bundle2.trk
```

```
$ dipy_horizon image.nii.gz
```

```
$ dipy_horizon tractogram.trk image.nii.gz --cluster --cluster_thr 15
```

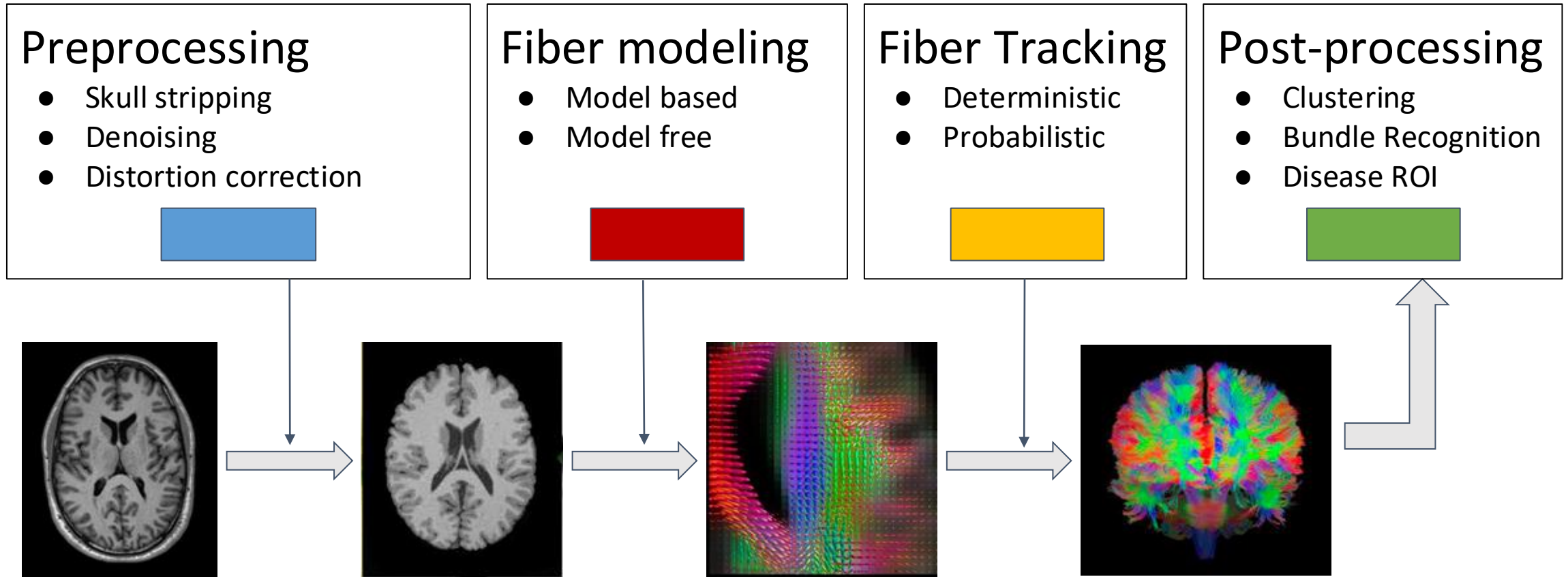
```
$ dipy_horizon bundle1.trk bundle2.trk --cluster --cluster_thr 15
```



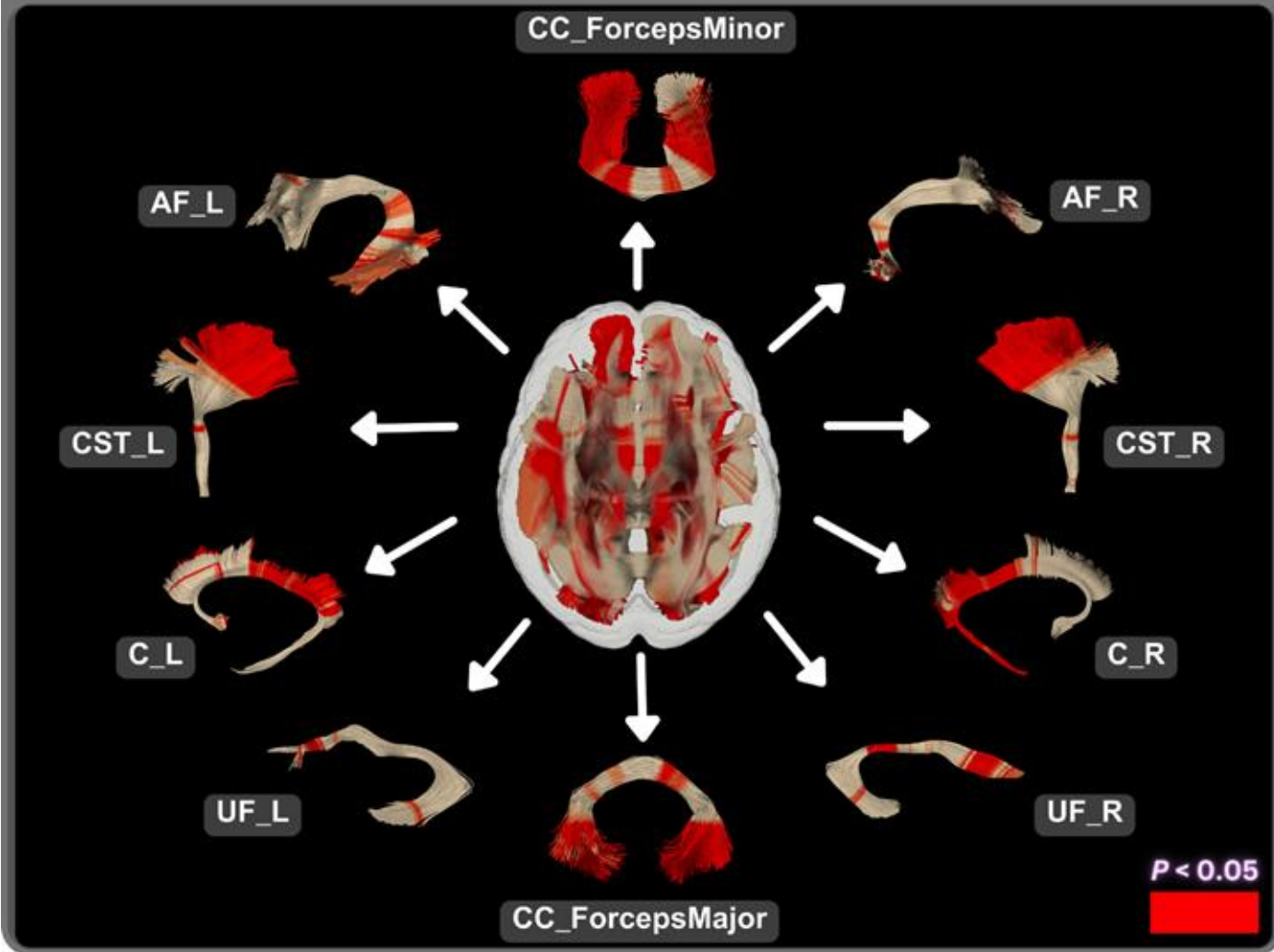
➔ Visualize your original data: `>>> dipy horizon my file.nii.gz`



Post-Processing: Clustering Analytics



Strongest Signal Detected by Tau and MD Metric



ADNI3 - 730 subjects

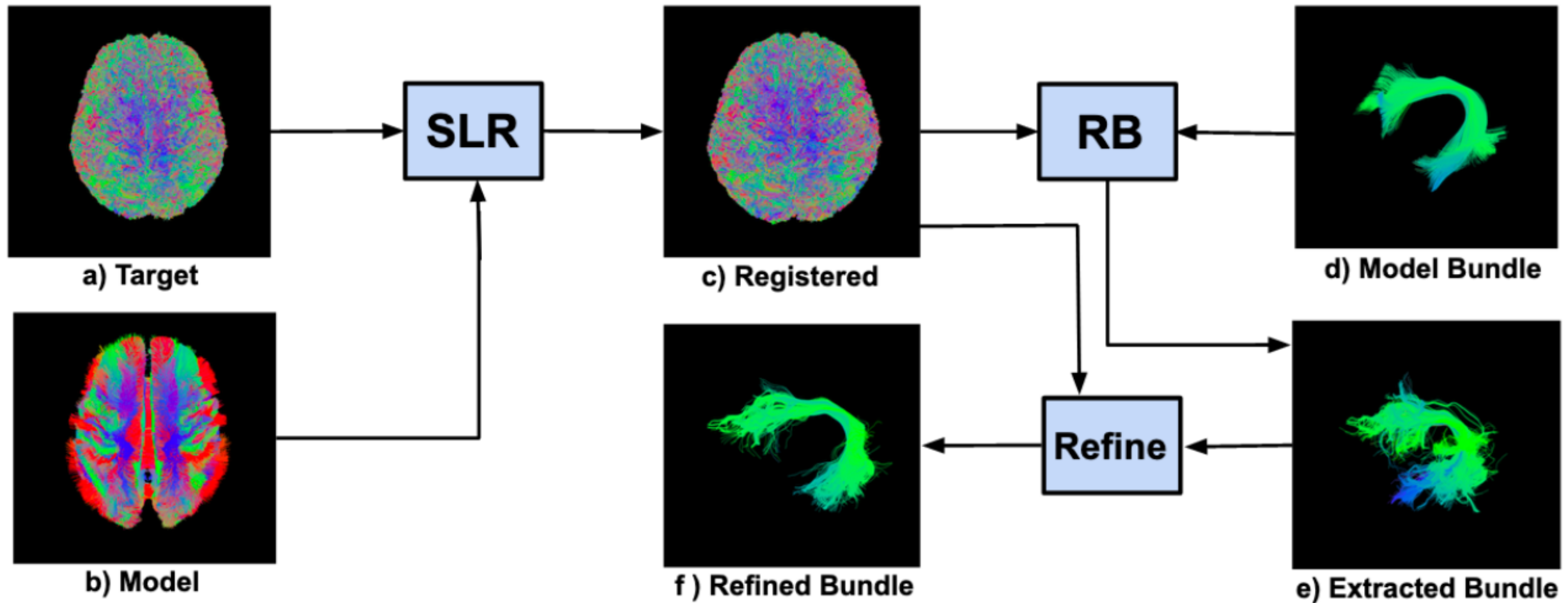
Chandio, B. Q., Villalon-Reina, J. E., Nir, T. M., Thomopoulos, S. I., Feng, Y., Benavidez, S., ... & Alzheimer's Disease Neuroimaging Initiative. (2024). Amyloid, Tau, and APOE in Alzheimer's Disease: Impact on White Matter Tracts. *PSB*, 2025



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➔ Recognize bundle



➔ Tractometry

Location

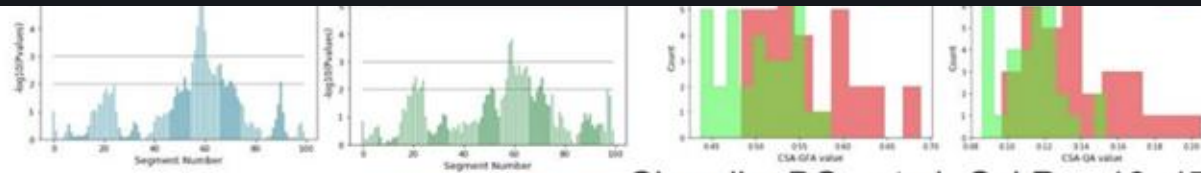
Significance

Detail

```
dipy_buan_profiles bundles/ subjects_small/ --out_dir "bundle_profiles"
```

```
dipy_buan_lmm "bundle_profiles/*" --out_dir "lmm_plots"
```

```
dipy_buan_shapes subjects_small/ --out_dir "sm_plots"
```



Chandio, BQ., et al. *Sci Rep* 10, 17149 (2020).



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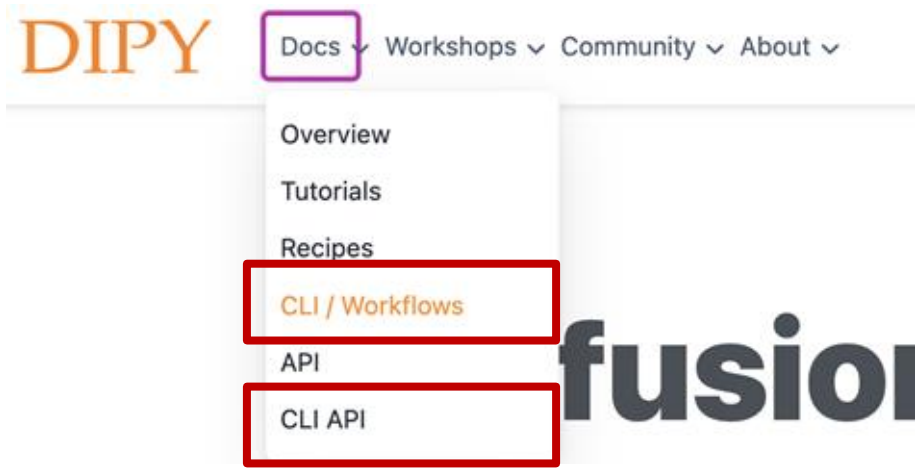
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Utilities

```
>>> dipy_fetch my_data|list|all
>>> dipy_math "vol1+vol2" data1.nii.gz data2.nii.gz
>>> dipy_extract_b0 my_data.nii.gz
>>> dipy_extract_shell my_data.nii.gz
>>> dipy_extract_volume my_data.nii.gz
>>> dipy_mask my_data.nii.gz 0.5
>>> dipy_convert
>>> dipy_concatenate_tractogram
>>> dipy_pam2nifti ....
>>> ....
```

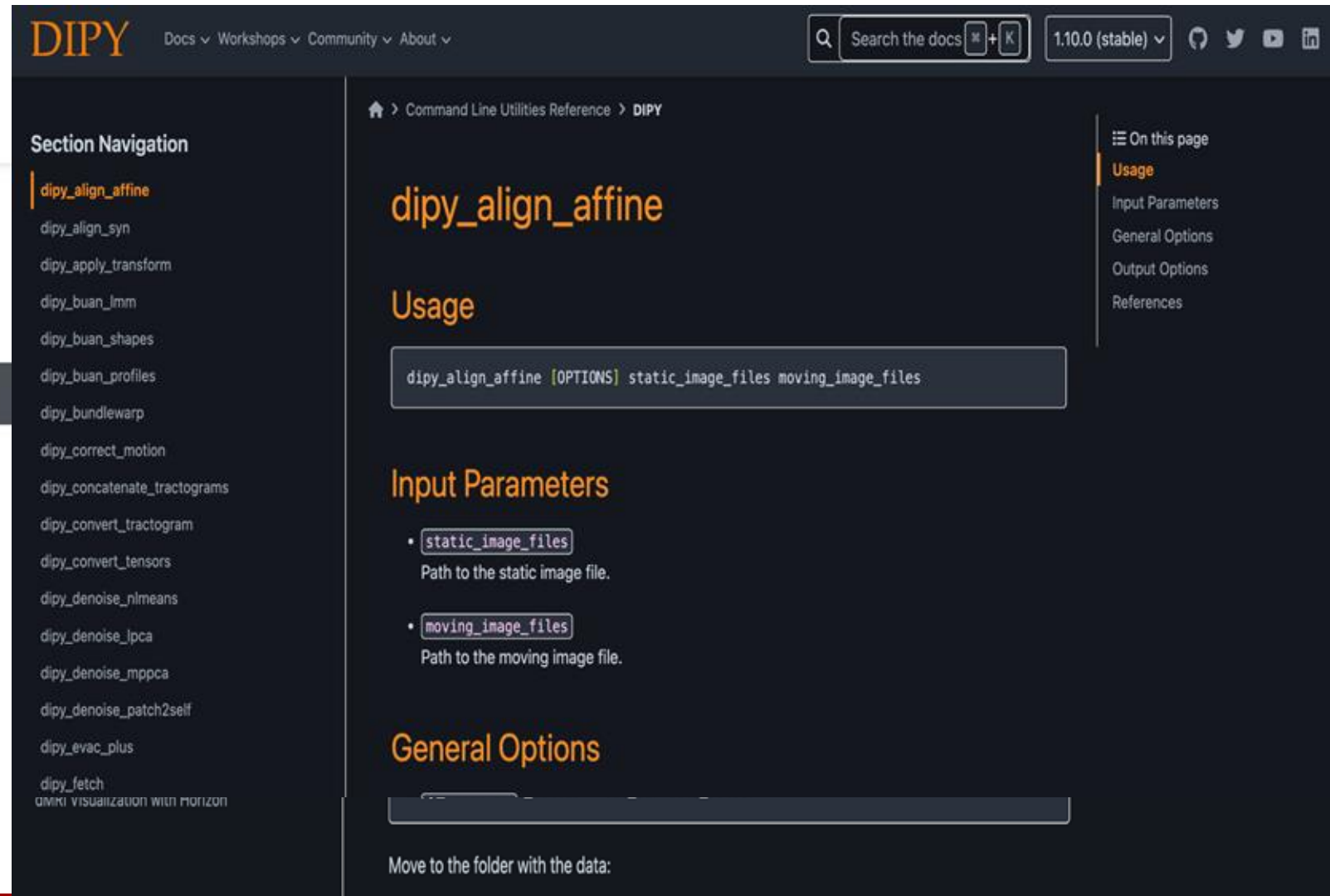


➡ Documentation & Tutorial



In a terminal: `$ dipy_` ←TAB

or eg. `dipy_info --help`



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What's Next ?



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➡ Continue to grow to achieve TRL 7-8 from our current status TRL 6.8

➡ To do so:

➡ Increase the number of DIPY Workflows

➡ Introduces basic User Interfaces, AI via Llm

➡ Increased validation against clinical gold standards

➡ BE PART of the COMMUNITY !!!

➡ By providing feedback of your clinical need/challenges

➡ By sharing data, building collaboration at every level



DIPY WORKSHOP 2027

March 15th – March 19th

Coming Soon

Join us for an immersive experience in the world of diffusion MRI.

Please enter your email

Notify Me

Notify me when the workshop registrations are available.

Follow Us



Why join the workshop?

Explore cutting-edge topics, network with researchers, and gain hands-on experience with DIPY tools.

"Thank you to all the amazing speakers who shared their expertise, and to every participant who brought curiosity and energy to the workshop. Your presence made it truly special. We're grateful to have had you with us!"

- DIPY Team

► Previous Workshops

Thank you for your attention.

DIPY

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